DRL-ENHANCED GSA-CEAR ADAPTIVE COMPRESSION & ENERGY AWARE-ROUTING FOR LIFETIME MAXIMIZATION IN UWSN

A project report submitted

In partial fulfillment for the award of the degree of

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CERTIFICATE

This is to certify that the project report entitled "DRL-ENHANCED GSA-CEAR ADAPTIVE COMPRESSION & ENERGY AWARE FOR ROUTING FOR LIFETIME MAXIMIZATION IN UWSN" is submitted by G. VANI, Y. THARUN, E. HASWANTH, S. L. CHAKRADHAR to the Y.S.R Engineering College of Yogi Vemana University, Proddatur, in partial fulfillment for the award of the degree of B. Tech in Computer Science and Engineering is a bonafide record of project work carried out by them under my supervision.

The contents of this report, in full or in parts, have not been submitted to any other Institution or University for the award of any degree or diploma.

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EXAMINERS:

INTERNAL EXAMINER **EXTERNAL EXAMINER**

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ABSTRACT

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Underwater Wireless Sensor Networks (UWSNs) have played an increasingly important role in the field of ocean environments for applications such as climate analysis, marine ecosystem observation, and monitoring offshore underwater structures. Underwater communication mainly uses acoustic signals instead of radio waves. This creates several challenges such as high signal attenuation, limited bandwidth, long propagation delay, and high error rates. As a result, sensor nodes experience rapid battery depletion, reducing network lifetime and coverage. Most UWSN routing and clustering methods select next-hop nodes or cluster heads, but do not consider redundant data, leading to unnecessary energy consumption during data transmission. In the proposed study, GSA-CEAR reduces transmitted data volume using correlation entropy-based feature selection and identifies an optimized routing path, while removing redundant or less informative sensor data. Lightweight differential compression sends only the difference between current and previous data. An energy-aware intelligent routing mechanism is used to select optimal transmission paths, improving energy efficiency and network stability. It incorporates an acoustic propagation model which is involved in the model's underwater signal behaviour, underwater energy consumption model estimates transmission cost, and zone-based clustering model divides the network into zones to distribute load evenly. These helps in preventing energy holes and improving scalability. GSA-CEAR performance results extend network lifetime, improves overall energy efficiency. The proposed results reduce the unnecessary data transmission and save energy with an increase in network lifetime. A Deep Reinforcement Learning (DRL) based routing module is integrated to learn energy-efficient forwarding paths using reward feedback derived from residual energy and transmission distance.

Keywords: Underwater wireless sensor networks (UWSNs), energy-efficient routing, deep reinforcement learning (DRL), data compression, cluster-based routing, network lifetime optimization.

CHAPTER 1

INTRODUCTION

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1.1 OVERVIEW

Underwater Wireless Sensor Networks (UWSNs) are specialized wireless sensor networks designed for monitoring and exploring aquatic environments. These networks consist of spatially distributed sensor nodes deployed underwater to collect and transmit data related to environmental conditions, marine life, underwater structures, and geological activities. UWSNs play a crucial role in applications such as oceanographic data collection, offshore oil exploration, pollution monitoring, disaster prevention, and military surveillance.

Unlike terrestrial wireless sensor networks, UWSNs operate in a highly challenging communication environment. The underwater medium presents unique physical characteristics such as high signal attenuation, limited bandwidth, long propagation delays, and dynamic topology changes caused by water currents. These factors significantly affect communication reliability and network performance.

Electromagnetic waves, commonly used in terrestrial wireless communication, suffer severe attenuation in water, especially in saline environments. Therefore, UWSNs primarily rely on acoustic wave propagation for communication. Acoustic signals can travel long distances underwater but introduce constraints such as low data rates, high latency, and susceptibility to noise and multipath interference.

In a typical UWSN architecture, underwater sensor nodes collect environmental data and transmit it to surface sink nodes or buoys. These surface stations forward the data to onshore control centers via satellite or radio communication for processing and analysis. Despite their advantages, UWSNs face critical challenges related to energy efficiency, reliable communication, and adaptive routing.

1.2 COMPONENTS OF UWSNs

An Underwater Wireless Sensor Network is composed of multiple interconnected components that collectively enable sensing, communication, and data transmission.

Sensor Nodes

Sensor nodes are the fundamental building blocks of UWSNs. They are equipped with sensing units to measure environmental parameters such as temperature, pressure, salinity, turbidity, dissolved oxygen, and chemical pollutants. Each node includes a processor, memory unit, battery power supply, and an acoustic modem for communication.

Acoustic Communication System

Since radio waves attenuate rapidly underwater, acoustic communication is used for data transmission. Acoustic waves can propagate over long distances underwater, making them suitable for underwater networking. However, acoustic communication suffers from low bandwidth, long propagation delay, Doppler shifts, and multipath fading.

Surface Buoys (Sink Nodes)

Surface buoys serve as gateways between underwater sensor nodes and terrestrial monitoring stations. They collect data from underwater nodes and transmit it to shore-based control centers using radio or satellite links. Surface buoys may also include GPS modules for position tracking and synchronization.

Energy Supply and Power Management

Underwater sensor nodes operate on limited battery power and are often deployed in inaccessible locations, making battery replacement impractical. Efficient power management and energy-aware communication strategies are essential to extend network lifetime.

Routing Protocols

Routing protocols determine how data packets travel from sensor nodes to the sink. Due to the dynamic underwater environment, routing protocols must be adaptive, energy-efficient, and robust against link failures and node mobility.

Localization System

Localization is essential for determining the positions of sensor nodes in a three-dimensional underwater space. Since GPS signals cannot penetrate water, alternative localization techniques such as Time of Arrival (ToA), Time Difference of Arrival (TDoA), and Angle of Arrival (AoA) are used.

Autonomous Underwater Vehicles (AUVs)

AUVs act as mobile nodes that assist in data collection, network maintenance, localization, and dynamic network reconfiguration. They are particularly useful in sparse deployment scenarios.

Data Aggregation and Processing Units

Data aggregation techniques reduce redundant transmissions by summarizing and compressing sensed data. This helps minimize communication overhead and conserves energy resources.

1.3 COMMUNICATION IN UWSNs

Communication in UWSNs is primarily achieved using acoustic waves due to the rapid attenuation of radio signals underwater. Acoustic communication enables long-range data transmission but introduces challenges such as limited bandwidth, high latency, and vulnerability to environmental noise and multipath propagation.

Underwater environments are highly dynamic, and sensor nodes may drift due to water currents. This mobility results in frequent topology changes and variable channel conditions. Doppler shifts caused by node movement further distort signal frequencies, affecting communication reliability.

To address these challenges, adaptive modulation techniques, error correction schemes, and signal processing methods are employed. In certain applications, hybrid communication systems combine acoustic and optical communication. Optical communication provides high-speed data transfer over short distances, while acoustic communication supports long-range transmission.

1.4 ISSUES IN UWSNs

Underwater wireless sensor networks face numerous technical challenges due to the harsh underwater environment. One of the primary issues is signal attenuation and absorption, which makes radio communication impractical and necessitates acoustic transmission with lower bandwidth and higher latency.

Multipath propagation occurs when signals reflect off the sea surface and seabed, causing interference and signal distortion. Environmental noise from marine life, ships, and natural disturbances further degrades communication quality.

Node mobility caused by water currents leads to frequent topology changes, complicating routing and synchronization. Energy constraints represent a major challenge since underwater nodes operate on limited battery power and replacing batteries is often impractical.

Other issues include Doppler effects, high bit error rates, and difficulty in accurate localization. These challenges necessitate the development of specialized communication protocols and energy-efficient routing strategies.

1.5 APPLICATIONS OF UWSNs

UWSNs have wide-ranging applications across scientific, industrial, and defense sectors.

- Environmental Monitoring: monitoring water quality, pollution levels, and marine ecosystems.
- Disaster Prevention: seismic monitoring for tsunami and earthquake detection.
- Military and Defense: underwater surveillance and intrusion detection.
- Industrial Monitoring: offshore pipeline monitoring and structural health assessment.
- Scientific Research: oceanographic data collection and deep-sea exploration.

1.6 FUTURE TRENDS IN UWSNs

Emerging technologies are driving the evolution of UWSNs toward greater autonomy and efficiency.

- Integration of Artificial Intelligence and Machine Learning for adaptive routing and decision-making.
- Energy harvesting techniques using ocean currents and renewable sources.
- Hybrid acoustic-optical communication systems.
- Blockchain-based security for secure underwater communication.
- These advancements aim to enhance reliability, efficiency, and scalability of underwater monitoring systems.

1.7 NEED FOR INTELLIGENT ROUTING IN UWSNs

Underwater Wireless Sensor Networks operate in highly dynamic and harsh environments where traditional routing protocols often fail to provide efficient communication. Acoustic communication suffers from long propagation delay, limited bandwidth, high attenuation, and frequent link failures. In addition, underwater sensor nodes are powered by limited battery energy, and replacing or recharging them is extremely difficult. These factors make routing in UWSNs a challenging task.

Conventional routing methods mainly focus on fixed path selection or basic cluster-based communication, but they do not adapt well to changing underwater conditions. As a result, data packets may be transmitted through inefficient paths, causing high energy consumption, packet loss, increased delay, and reduced network lifetime.

To overcome these limitations, intelligent routing mechanisms are required in UWSNs. Intelligent routing can dynamically select the best forwarding path based on parameters such as residual energy, transmission distance, link quality, node density, and network congestion. By making adaptive routing decisions, the system can improve energy efficiency, reduce unnecessary retransmissions, and increase the reliability of underwater communication.

Therefore, the use of optimization and machine learning-based routing approaches has become important in modern UWSNs for achieving stable, energy-aware, and long-lasting network performance.

1.8 ROLE OF MACHINE LEARNING IN UWSNs

Machine learning techniques enable intelligent decision-making and adaptive optimization in underwater sensor networks. ML improves routing efficiency, energy management, fault detection, and communication reliability.

ML-based routing protocols can select optimal paths based on network conditions, reducing packet loss and conserving energy. Data compression and aggregation techniques help minimize redundant transmissions. Machine learning also enhances anomaly detection, predictive maintenance, and network security. By enabling adaptive and data-driven optimization, ML significantly improves the efficiency and reliability of UWSNs.

The primary purpose of applying machine learning in this project is to achieve adaptive energy-aware routing. In underwater environments, sensor nodes are battery-powered, and replacing or recharging them is extremely difficult and expensive. Deep Reinforcement Learning enables the system to learn optimal routing policies by interacting with the environment and continuously improving its decisions based on feedback such as residual energy, link quality, node distance, and congestion levels. This ensures that data packets are transmitted through the most energy-efficient and stable paths, thereby significantly extending the overall network lifetime.

Machine learning also enhances network stability and reliability. In UWSNs, node mobility and environmental disturbances frequently cause link failures. DRL models continuously monitor network parameters and predict optimal forwarding strategies to avoid unstable or energy-depleted nodes. This reduces packet loss, improves delivery ratio, and minimizes retransmissions.

1.9 TYPES OF MACHINE LEARNING

Machine Learning (ML) is a subset of artificial intelligence (AI) that enables systems to learn patterns from data and make decisions or predictions without explicit programming. ML is broadly categorized into three main types: Supervised Learning, Unsupervised Learning, and Reinforcement Learning. Additionally, there are subcategories and hybrid approaches that extend the capabilities of these primary types. Understanding these categories helps in selecting the appropriate ML technique for a given application.

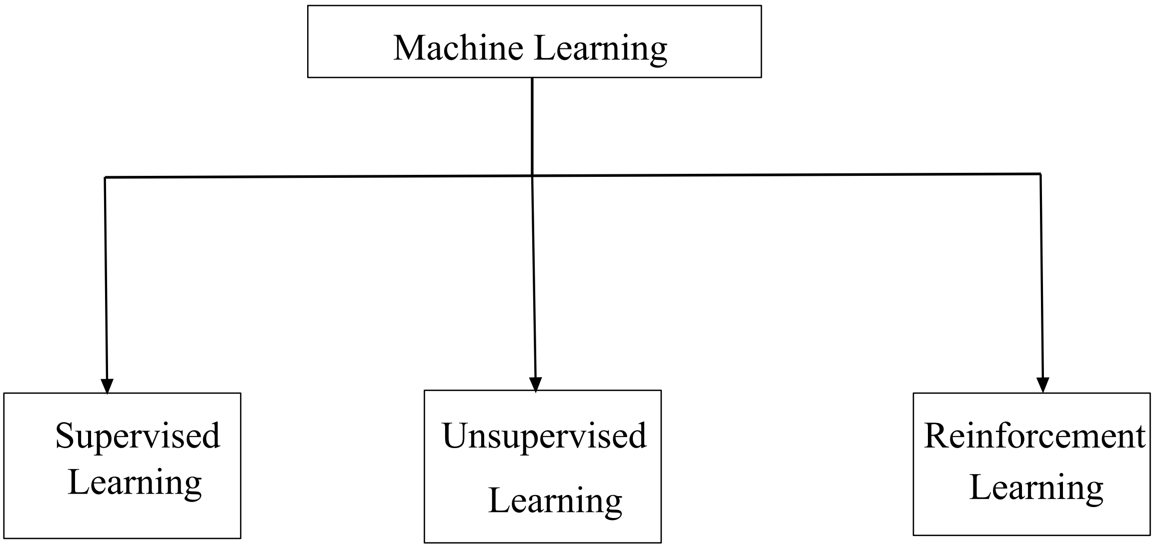

Fig1.1 Types of Machine Learning

Supervised Learning

Supervised learning is a type of ML where the model is trained using labelled data. The algorithm learns from input-output pairs and maps the input to the desired output based on patterns found in the data.

Key Characteristics:

- Requires labelled datasets.
- Used for classification and regression problems.
- Model learns from historical data and generalizes to new data.

Common Algorithms:

- Linear Regression: Used for predicting continuous values (e.g., house price prediction).
- Logistic Regression: Used for binary classification problems (e.g., spam detection). Support Vector Machines (SVMs): Used for classification by finding the best decision boundary.
- Decision Trees: Hierarchical model that splits data based on features.
- Random Forest: An ensemble of decision trees that improves prediction accuracy.
- Neural Networks: Used for complex pattern recognition, such as image and speech recognition.

Applications:

- Medical diagnosis (e.g., detecting diseases from medical images).
- Fraud detection in financial transactions.
- Sentiment analysis in customer reviews.
- Speech and image recognition.

Unsupervised Learning

Unsupervised learning involves training a model using data that is not labelled. The algorithm identifies patterns, structures, or groupings in the data without explicit supervision.

Key Characteristics:

- Works with unlabelled data.
- Used for clustering, association, and dimensionality reduction tasks.
- Finds hidden patterns and anomalies in data.

Common Algorithms:

- K-Means Clustering: Groups data points into k clusters based on similarity.
- Hierarchical Clustering: Creates a hierarchy of nested clusters.
- Principal Component Analysis (PCA): Reduces the dimensionality of data while preserving key features.
- Autoencoders: Neural networks used for unsupervised feature learning and anomaly detection.
- Gaussian Mixture Models (GMMs): Probabilistic model used for soft clustering.

Applications:

- Market segmentation for targeted advertising.
- Anomaly detection in network security.
- Recommender systems (e.g., movie or product recommendations).
- Image compression and feature extraction.

Reinforcement Learning

Reinforcement Learning (RL) is a type of ML where an agent interacts with an environment to learn optimal behaviours through rewards and punishments.

Key Characteristics:

- No labelled data; learns by trial and error.
- Uses reward-based learning to optimize decisions.
- Consists of agents, actions, states, and rewards.

Common Algorithms:

- Q-Learning: A value-based approach that finds the best action for each state.
- Deep Q-Networks (DQN): Uses deep learning to approximate Q-values.
- Policy Gradient Methods: Optimizes policies directly rather than estimating value functions.

Applications:

- Game playing (e.g., AlphaGo, OpenAI's Dota bot).
- Robotics and autonomous control systems.
- Traffic signal optimization.
- Portfolio management in finance.

Machine Learning encompasses a broad range of approaches, each suited for different types of problems. Supervised learning is best for structured prediction tasks, unsupervised learning excels in pattern discovery, and reinforcement learning is ideal for sequential decision-making. Hybrid methods like semi-supervised learning and transfer learning provide flexibility, while deep learning enables breakthroughs in AI-driven applications.

Understanding these ML types helps in selecting the best approach for solving complex real-world problems, from medical diagnosis to autonomous systems and beyond.

1.10 PROBLEM STATEMENT

Underwater Wireless Sensor Networks (UWSNs) are widely used for marine monitoring, oceanographic data collection, disaster prevention, and underwater surveillance. However, underwater communication is difficult because acoustic signals suffer from high propagation delay, limited bandwidth, attenuation, and high energy consumption. These factors reduce network efficiency and lifetime.

The objectives of this project are as follows:

➢ To remove redundant underwater sensor data using feature selection.

➢ To reduce transmission size using lightweight compression.

➢ To optimize cluster formation using GSA.

➢ To perform adaptive routing using DRL.

➢ To improve network lifetime and energy efficiency.

Traditional routing methods mainly focus on path selection or cluster-head formation, but they do not effectively handle redundant data and dynamic underwater conditions. As a result, the network suffers from high energy consumption, packet loss, increased delay, and early node death. Therefore, there is a need for an intelligent and integrated framework such as DRL-Enhanced GSA-CEAR to improve routing efficiency and maximize network lifetime in UWSNs.

1.11 MOTIVATION

Underwater Wireless Sensor Networks (UWSNs) are an important technology used in applications such as marine environmental monitoring, oceanographic data collection, disaster prevention, underwater surveillance, and offshore infrastructure inspection. However, the performance of UWSNs is greatly affected by the harsh underwater environment, where communication mainly depends on acoustic signals. These signals suffer from high propagation delay, limited bandwidth, severe attenuation, multipath fading, and high energy consumption. As a result, underwater sensor nodes lose energy quickly, communication becomes unreliable, and the overall network lifetime is reduced. The primary motivation of this project is to develop an intelligent and energy-efficient routing framework that can reduce unnecessary data transmission, improve communication efficiency, and extend the operational lifetime of UWSNs.

This project also aims to modernize underwater communication systems by integrating advanced optimization and learning techniques into routing and data transmission. Most existing routing protocols mainly focus on path selection or cluster-head formation, but they often ignore the transmission of redundant sensor data, which leads to extra communication overhead and faster battery depletion. By combining correlation–entropy-based feature selection, lightweight differential compression, Gravitational Search Algorithm (GSA) based clustering, and Deep Reinforcement Learning (DRL) based routing, the proposed model can reduce data redundancy, optimize cluster formation, and select efficient routing paths adaptively. This helps in balancing energy consumption, improving packet delivery ratio, reducing delay, and supporting long-term underwater monitoring applications.

Finally, the project seeks to demonstrate the practical benefits of artificial intelligence and optimization techniques in Underwater Wireless Sensor Networks. By providing a reliable, adaptive, and energy-aware communication framework, it encourages the use of intelligent technologies in underwater monitoring and marine research applications. It also creates a foundation for future improvements in smart underwater communication systems and contributes to the development of efficient, scalable, and sustainable UWSN solutions.

CHAPTER 2

LITERATURE SURVEY

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LITERATURE SURVEY

2.1 INTORDUCTION TO LITERATURE SURVEY

Literature Survey is the process of summarizing the research work done by various researchers in a specific area of study. In this paper, various works carried out by the researchers based on Underwater Wireless Sensor Networks (UWSNs), including routing protocols, clustering techniques, data compression methods, and machine learning-based approaches, are discussed. It will help us to understand the techniques used, advantages, disadvantages, scope, and gaps of the existing research work in the field of energy-efficient routing and communication in UWSNs.

Here, different approaches such as depth-based routing, clustering-based routing, optimization techniques like GSA, DQN, and Reinforcement Learning (RL) are studied to find out the best model for improving network lifetime and energy efficiency in underwater communication. This chapter discusses the literature survey on various routing strategies, feature selection methods, clustering techniques, optimization methods, and evaluation metrics used in different studies. In addition, the shortcomings of the various studies carried out in this area are also given. The shortcomings include high energy consumption, redundant data transmission, poor adaptability to underwater conditions, high computational cost, and limited scalability. These shortcomings form the basis for proposing an efficient and reliable routing model in this study.

2.2 LITERATURE SURVEY

Ala Al-Fuqaha [1] et al., **"Internet of Undersea Things (IoUT): Recent Advances and Future Directions,"** IEEE Internet of Things Journal, 2022". The authors presented a detailed survey on the Internet of Undersea Things (IoUT) and its role in advancing underwater communication technologies. The paper explains how underwater sensors, Autonomous Underwater Vehicles (AUVs), and smart monitoring devices are interconnected using acoustic, optical, and magnetic induction communication methods. The study highlighted important applications such as marine environment monitoring, oil and gas exploration, underwater surveillance, climate research, and disaster prediction. It also discussed key challenges such as high energy consumption, long propagation delay, low bandwidth, node mobility, and harsh underwater channel conditions. The paper suggested that future UWSN systems can be improved using intelligent routing protocols, energy-efficient communication strategies, and artificial intelligence techniques.

M. Stojanovic and P. A. van Walree [2] et al., **“Underwater Acoustic Communication Channels: Propagation Models and Performance,”** 2021. The authors focused on the fundamental characteristics of underwater acoustic communication channels. The paper explains how sound waves are used for underwater communication since radio and optical signals have severe limitations in water. It discussed different propagation models by considering factors such as attenuation, multipath fading, Doppler spread, ambient noise, and long propagation delay. The study also analyzed how temperature, salinity, water depth, and surface motion affect channel performance. The paper emphasized the need for adaptive modulation, equalization, and energy-efficient transmission methods to improve underwater communication reliability.

N. Saeed, A. Celik, T. Y. Al-Naffouri, and M. Alouini [3] et al., **“Underwater Optical Wireless Communications, Networking, and Localization: A Survey,”** Ad Hoc Networks, 2020. The authors presented a comprehensive survey on Underwater Optical Wireless Communication (UOWC) as an emerging alternative to acoustic communication. The study highlighted the advantages of optical communication such as high data rate, low latency, and enhanced security. However, it also discussed major challenges including limited transmission range, absorption and scattering effects, alignment issues, and sensitivity to water turbidity. The paper further explored networking architectures, routing protocols, localization techniques, and hybrid acoustic-optical systems. The work is useful for understanding next-generation underwater communication technologies.

Z. Luo and M. Zuba [4] et al., **“Localization Challenges in Underwater Sensor Networks,”** 2018. The authors analyzed localization techniques and challenges in Underwater Sensor Networks (UWSNs). The paper explained that accurate localization is essential for data collection, routing, and underwater monitoring applications. Unlike terrestrial networks, underwater systems cannot use GPS, and they face issues such as long propagation delay, limited bandwidth, node mobility, multipath fading, and high energy consumption. The study categorized localization methods into range-based and range-free approaches, including Time of Arrival (ToA), Time Difference of Arrival (TDoA), and Received Signal Strength (RSS)-based methods. It also discussed future directions such as energy-efficient localization and hybrid acoustic-optical solutions.

Muhammad Nadeem, Nadeem Javaid, Muhammad Imran, and Mohsen Guizani [5] et al., **"Improved Depth-Based Routing Protocol for Underwater Wireless Sensor Networks,"** 2021. The authors proposed an enhanced Depth-Based Routing (DBR) protocol to improve the performance of Underwater Wireless Sensor Networks. Traditional DBR protocols rely on depth information to forward packets, but they often suffer from redundant transmissions, packet loss, and high energy consumption. The proposed protocol optimized forwarding decisions using depth difference, residual energy, and controlled holding time. Simulation results showed that the improved DBR protocol achieved better energy efficiency, lower end-to-end delay, improved packet delivery ratio, and longer network lifetime compared to conventional DBR methods.

Natchapon Chirdchoo, Wee-Seng Soh, and Kee-Chaing Chua [6] et al., **"A Survey of Routing Protocols for Underwater Wireless Sensor Networks,"** 2020. The authors presented a detailed survey of routing protocols designed for Underwater Wireless Sensor Networks. The paper categorized routing protocols into depth-based, location-based, cluster-based, opportunistic, and energy-efficient routing schemes. It compared these protocols based on metrics such as packet delivery ratio, energy consumption, end-to-end delay, and network lifetime. The study also highlighted key design challenges including scalability, reliability, node mobility, and harsh underwater channel conditions. This paper provides a strong foundation for understanding existing routing strategies in UWSNs.

Paolo Casari and Michele Zorzi [7] et al., **"Protocols for Underwater Acoustic Networks: A Comparative Study,"** 2020. The authors presented a comparative study of communication protocols for underwater acoustic networks. The paper explained that traditional terrestrial communication protocols are not suitable for underwater systems because of long propagation delays, limited bandwidth, high bit error rates, and energy limitations. The study compared various MAC and routing protocols based on throughput, latency, reliability, and energy efficiency. It also discussed protocol trade-offs and adaptive strategies required to improve underwater communication performance.

Muhammad Ayaz, Imran Baig, Abdullah and Issa Faye [8] et al., **"Routing Techniques in Underwater Wireless Sensor Networks: A Survey,"** 2021. The authors examined different routing techniques developed for Underwater Wireless Sensor Networks. The paper categorized routing protocols into location-based, depth-based, cluster-based, opportunistic, energy-efficient, and bio-inspired routing approaches. It evaluated these protocols using metrics such as packet delivery ratio, end-to-end delay, energy efficiency, and network lifetime. The paper also identified major research challenges such as scalability, reliability, 3D deployment, and cross-layer optimization.

Hui Yan, Zhijie Shi, and Jun-Hong Cui [9] et al., **"DBR Routing Improvements for Underwater Networks,"** 2008. The authors proposed enhancements to the Depth-Based Routing (DBR) protocol for underwater networks. The original DBR protocol forwards packets based on node depth, but it suffers from redundant transmissions, high energy consumption, and increased packet collisions. The improved DBR introduced optimized packet forwarding decisions by adjusting holding time and depth thresholds. The modifications significantly improved packet delivery ratio, reduced delay, and extended network lifetime in underwater acoustic environments.

Mubashir Ahmed, Shafiullah Khan, and Jaime Lloret [10] et al**., "Energy-Efficient 3D Routing Protocols in Underwater Wireless Sensor Networks,"** 2022. The authors investigated energy-efficient routing strategies for three-dimensional (3D) underwater sensor networks. Since underwater nodes are deployed at different depths and move due to water currents, routing becomes more complex than in terrestrial networks. The proposed routing mechanisms considered depth, residual energy, distance to sink, and link quality to improve performance. Simulation results showed better reliability, lower delay, and improved energy efficiency compared to existing methods.

Myung K. Park, John Heidemann, and Wei Ye [11] et al., **"Energy Consumption Modeling for Underwater Acoustic Sensor Networks,"** 2020. The authors presented a detailed energy consumption model for underwater acoustic sensor networks. The study analyzed how energy is consumed during transmission, reception, idle listening, and signal processing in acoustic modems. It developed mathematical models to estimate energy usage based on distance, packet size, data rate, and channel conditions. The work helps in accurately evaluating routing protocols and predicting network lifetime in underwater environments.

Roee Diamant and Lutz Lampe [12] et al., **"Underwater Localization: Algorithms and Performance Analysis,"** 2011. The authors proposed localization algorithms for underwater acoustic sensor networks, where GPS cannot be used. The paper explained how acoustic measurements such as Time of Arrival (ToA) and Time Difference of Arrival

(TDoA) can be used to estimate node positions. It also studied the effects of multipath propagation, synchronization errors, and limited bandwidth on localization performance. The results showed that accurate localization is important for improving routing and communication efficiency in UWSNs.

Mari Carmen Domingo [13] et al., **“Overview of Channel Models for Underwater Wireless Communication Networks,”** 2020. The author presented an overview of underwater wireless communication channel models. The paper discussed how underwater acoustic channels are affected by attenuation, multipath propagation, Doppler spread, ambient noise, and long delays. It also reviewed path loss, noise, and fading models used to characterize underwater communication. This work is important for designing reliable routing protocols and energy-efficient communication strategies in underwater networks.

R. Shankar [14] et al., **“Attenuation Modelling for Underwater Acoustic Networks,”** 2020. The authors focused on signal attenuation modeling in underwater acoustic networks. The study explained how acoustic signals lose strength due to spreading loss, absorption, scattering, and environmental factors such as temperature, salinity, depth, and frequency. The proposed attenuation models help estimate communication range, required transmission power, and link reliability. This work is important for designing energy-efficient routing mechanisms in UWSNs.

Melike Erol-Kantarci, Hussein T. Mouftah, and Sema Oktug [15] et al., **“Underwater Wireless Sensor Networks: Communication Architectures and Protocols,”** 2021. The authors provided a comprehensive survey of communication architectures and protocols for Underwater Wireless Sensor Networks. The paper discussed centralized, distributed, and hierarchical network structures, along with acoustic, optical, and hybrid communication methods. It also examined MAC, routing, and transport layer protocols tailored for underwater environments. The study highlighted key issues such as energy constraints, node mobility, high delay, and harsh channel conditions.

S. Khan [16] et al., **“Vector-Based Forwarding Protocols for Underwater Wireless Sensor Networks: Performance Evaluation,”** 2020. The authors evaluated Vector-Based Forwarding (VBF) protocols for Underwater Wireless Sensor Networks. VBF is a geographic routing approach that forwards packets along a virtual routing vector toward the sink node. The paper analyzed its performance in terms of packet delivery ratio, end-to-end delay, energy consumption, and network lifetime. Results showed that VBF helps reduce unnecessary transmissions in dense underwater networks while maintaining reliable communication.

Abdul Wahid, Sang-Ha Lee, and Dongkyun Kim [17] et al., **“Energy Efficient Routing Protocol for Underwater Wireless Sensor Networks,”** 2012. The authors proposed an energy-efficient routing protocol for UWSNs that selects forwarding nodes based on depth and residual energy. The protocol minimized unnecessary transmissions and balanced energy consumption among nodes. Performance evaluation showed lower energy usage, higher reliability, and improved network lifetime compared to traditional underwater routing schemes.

Nadeem Javaid [18] et al., **“EECMR Clustering Protocol for Underwater Networks,”** 2019. The authors introduced the Energy-Efficient Clustered Multi-hop Routing (EECMR) protocol for underwater networks. The protocol organized sensor nodes into clusters, where cluster heads were selected based on residual energy, node depth, and distance to the sink. The multi-hop communication approach reduced long-distance transmissions and improved energy efficiency. Simulation results showed better packet delivery ratio, lower delay, and extended network lifetime.

Kejie Chen [19] et al., **“HydroCast: Pressure Routing for Underwater Sensor Networks,”** 2012. The authors proposed HydroCast, a pressure-based geographic routing protocol for underwater sensor networks. Instead of using full location information, HydroCast used pressure (depth) values to forward packets toward the surface sink. The protocol also introduced a recovery mechanism to handle communication voids where no suitable forwarding node is available. Results showed improved packet delivery ratio, lower delay, and better energy efficiency compared to existing protocols.

A. Qureshi [20] et al., **“Energy Consumption Analysis in Underwater Wireless Sensor Networks,”** 2020. The authors analyzed energy consumption patterns in UWSNs by studying the impact of transmission, reception, idle listening, sensing, and packet retransmissions. The paper also considered underwater acoustic channel characteristics such as attenuation, delay, and error rates. It highlighted the importance of adaptive power control, efficient MAC protocols, and energy-aware routing to extend network lifetime.

A. Singh and R. Sharma [21] et al., **"Energy-Efficient Clustering Techniques in Wireless Sensor Networks: A Review,"** 2020. The authors reviewed different clustering techniques developed for Wireless Sensor Networks (WSNs). The paper discussed cluster formation, cluster head selection, and data aggregation methods based on parameters such as residual energy, node degree, and distance. Although the study focused on terrestrial WSNs, the discussed clustering strategies are highly useful for UWSN projects, especially in designing scalable and energy-aware routing protocols.

Paolo Casari and Michele Zorzi [22] et al., **"Underwater Acoustic Communications: Channel Modeling and Protocol Design,"** 2020. The authors provided an in-depth overview of underwater acoustic communication systems, focusing on channel modeling and protocol design. The paper discussed high attenuation, limited bandwidth, long delays, Doppler effects, and multipath fading in underwater channels. It also examined MAC, routing, and cross-layer protocol design considerations. The study emphasized the need for adaptive, delay-tolerant, and energy-efficient communication protocols for reliable underwater networking.

2.3 SUMMARY OF LITERATURE SURVEY

The literature review of the proposed work identified many studies focused on improving Underwater Wireless Sensor Networks (UWSNs) using routing protocols, clustering techniques, optimization algorithms, and intelligent learning-based methods. Early works mainly focused on depth-based, location-based, and pressure-based routing, while recent studies explored GSA, DQN, RL, clustering, and energy-aware routing to improve packet delivery ratio, throughput, and network lifetime.

Although these methods achieved good results, several limitations still exist. Most existing approaches mainly focus on routing or clustering, but they do not effectively handle redundant data transmission, which increases energy consumption. In addition, many learning-based models have high computational cost, poor adaptability, and limited scalability in real underwater environments.

Therefore, there is a need for a more integrated and energy-efficient solution. Based on these limitations, the proposed DRL-Enhanced GSA-CEAR model combines feature selection, lightweight compression, GSA-based clustering, and DRL-based routing to reduce communication overhead and improve network lifetime in UWSNs.

CHAPTER 3

EXISTING METHOD

CHAPTER 3
EXISTING METHOD

3.1 CLUSTER-BASED ROUTING IN UNDERWATER WIRELESS SENSOR NETWORKS

3.1.1 Overview

Cluster-based routing is a hierarchical communication approach in which sensor nodes are grouped into clusters based on certain criteria such as distance, depth, or energy levels. Each cluster contains a cluster head and several member nodes. The member nodes sense environmental data and transmit it to their respective cluster head, while the cluster head performs data aggregation and forwards the processed data to the sink node.

The key advantage of this approach is that it reduces redundant data transmission and limits long-distance communication, which is the primary cause of energy depletion in UWSNs. Instead of every node sending data directly to the sink, only cluster heads are responsible for long-range communication, thereby conserving energy for regular sensor nodes.

In most clustering techniques, the role of the cluster head is not fixed and is rotated among nodes based on factors such as residual energy, node density, or network conditions. This rotation helps in balancing the energy consumption across the network and prolongs the overall network lifetime.

Furthermore, cluster-based routing supports both single-hop and multi-hop communication. In single-hop communication, cluster heads directly transmit data to the sink, whereas in multi-hop communication, data is forwarded through intermediate cluster heads. Multi-hop communication is more suitable for large-scale UWSNs where the distance between nodes and the sink is significant.

Overall, cluster-based routing provides an efficient framework for managing communication in underwater environments by improving energy efficiency, reducing network congestion, and enhancing data transmission reliability.

3.2 INTRODUCTION

In recent years, Underwater Wireless Sensor Networks (UWSNs) have attracted significant attention due to their wide range of applications in ocean monitoring, underwater exploration, military surveillance, disaster prevention, and environmental data collection. UWSNs consist of multiple sensor nodes deployed underwater to sense physical parameters such as temperature, pressure, salinity, and movement of underwater objects. These nodes communicate with each other and transmit the collected data to a sink node located near or above the water surface. However, underwater communication is highly challenging because it mainly relies on acoustic signals, which suffer from high attenuation, limited bandwidth, long propagation delay, multipath fading, and high energy consumption.

One of the major issues in UWSNs is the limited battery power of sensor nodes, since replacing or recharging batteries underwater is difficult and costly. Due to this reason, energy efficiency and network lifetime maximization are considered as important research problems in underwater communication. Traditional routing methods often fail to provide efficient data transmission because they do not effectively balance the communication load among sensor nodes. As a result, some nodes lose energy quickly, causing early node death and reduced network lifetime.

To overcome these limitations, several cluster-based routing protocols have been proposed for UWSNs. In clustering, the underwater sensor nodes are grouped into clusters, and one node in each cluster is selected as a Cluster Head (CH). The cluster head collects data from the member sensor nodes, aggregates it, and forwards it to the sink node either directly or through multi-hop communication. This clustering mechanism helps in reducing unnecessary transmissions and improves energy utilization.

The existing method uses the Gravitational Search Algorithm (GSA) for cluster head selection and routing optimization in underwater wireless sensor networks. The network is divided into zones, where cluster heads are chosen based on residual energy, node density, and distance to the sink node. Sensor nodes transmit data to their cluster heads, which then forward aggregated data through energy-efficient routes to the sink. This approach improves load balancing, reduces transmission energy consumption, and extends network lifetime.

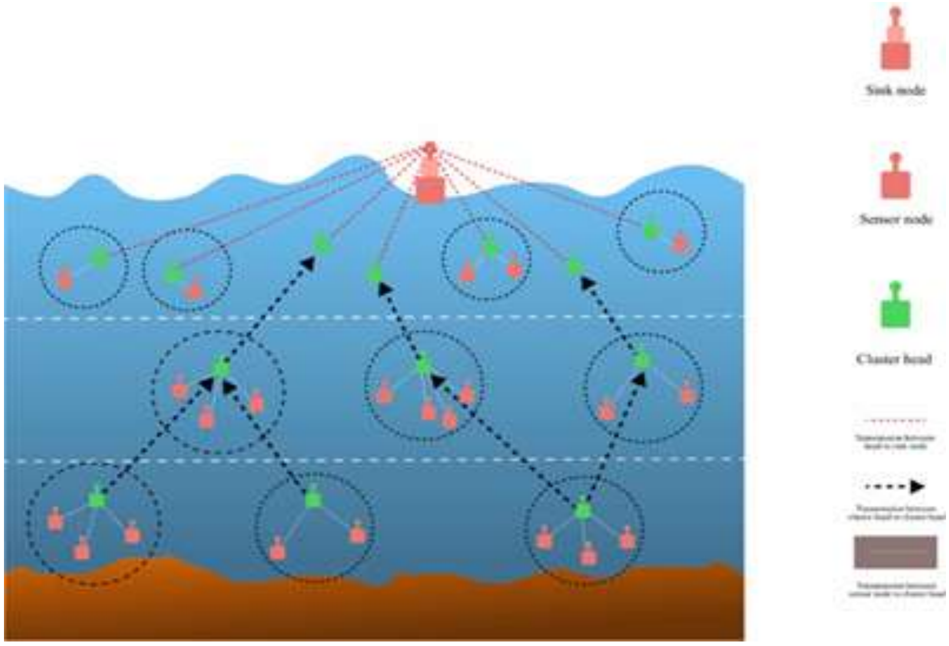

Fig 3.1 Scenario of underwater sensor network

The existing system architecture is designed for an underwater wireless sensor network using cluster-based routing. Sensor nodes are randomly deployed at different depth levels to monitor parameters such as temperature, pressure, salinity, and other oceanographic data. These nodes are organized into clusters, where one node is selected as the Cluster Head (CH) and the others act as ordinary sensor nodes. Cluster head selection is performed using the Gravitational Search Algorithm based on residual energy, node density, and distance.

The communication process in this architecture takes place in three levels:

1. Sensor Node to Cluster Head Communication

Each ordinary sensor node senses the underwater data and sends the collected information to its nearest cluster head. This is represented in the figure by the light blue communication links inside each cluster. This communication is called intra-cluster communication.

2. Cluster Head to Cluster Head Communication

After receiving the data from member nodes, the cluster head performs data aggregation to remove redundant information and reduce communication overhead. Then, instead of sending the data directly to the sink node, the cluster heads forward the aggregated data to other nearby cluster heads through multi-hop communication. This is shown by the black dashed arrows in the figure. This stage helps in reducing transmission energy, especially for nodes that are far away from the sink node.

3. Cluster Head to Sink Node Communication

Finally, the data is transmitted from the upper-level cluster heads to the sink node, which is located above the water surface. The sink node acts as the central data collection point and receives all the processed information from the underwater network. This communication is shown in the figure by the red dotted lines.

The underwater environment is divided into depth layers to support efficient clustering and routing. Nodes at lower depths forward data through upper-level cluster heads, reducing long-distance transmissions. This layered cluster-based architecture improves energy efficiency, load balancing, and network lifetime. Overall, by combining clustering, data aggregation, multi-hop routing, and Gravitational Search Algorithm-based cluster head selection, the network minimizes energy consumption and extends operational lifetime.

3.3 EXISTING ROUTING PROTOCOLS IN UWSN

Routing in underwater wireless sensor networks is essential for transmitting data efficiently from sensor nodes to the sink node. Due to underwater challenges such as high propagation delay, limited bandwidth, node mobility, energy constraints, and unreliable links, routing protocols must be adaptive and energy-aware. Unlike terrestrial networks, underwater routing requires specialized mechanisms to ensure reliable communication.

UWSN routing protocols are mainly classified into localization-based and localization-free approaches. Localization-based protocols use node position information to select optimal transmission paths, while localization-free protocols rely on factors such as hop count, depth, or connectivity. Localization-based methods improve routing efficiency, whereas localization-free methods are more practical where accurate positioning is difficult. Both approaches have benefits and limitations depending on network conditions and application needs.

Both approaches have their advantages and limitations, and their effectiveness depends on the network conditions and application requirements.

3.3.1 Localization-Based Routing Protocols

Localization-based routing protocols use the geographical position of sensor nodes to make routing decisions. In these protocols, each node is aware of its own location as well as the position of neighboring nodes or the sink node. This positional information helps in selecting the shortest or most efficient path for data transmission.

Localization in UWSNs is achieved using techniques such as Time of Arrival (ToA), Time Difference of Arrival (TDoA), and Received Signal Strength Indicator (RSSI). These methods estimate the position of nodes based on signal propagation characteristics. Once the location is determined, routing decisions are made by selecting nodes that are closer to the destination, thereby reducing transmission distance and energy consumption.

The main advantage of localization-based routing is that it enables more accurate and efficient data forwarding, leading to reduced delay and improved network performance. It also helps in avoiding unnecessary transmissions and reduces packet loss by choosing optimal paths.

However, these protocols have certain limitations. Obtaining accurate location information in underwater environments is difficult due to signal distortion, noise, and dynamic conditions. Localization techniques may require additional hardware or complex computations, which increase energy consumption. Therefore, while localization-based routing provides better accuracy, it may not always be feasible in practical underwater scenarios.

3.3.2 Localization-Free Routing Protocols

Localization-free routing protocols do not depend on the exact geographical position of sensor nodes. Instead, they rely on alternative parameters such as hop count, node connectivity, depth information, or residual energy to make routing decisions. These protocols are widely used in UWSNs because they are simpler and more energy-efficient compared to localization-based methods.

One common approach in localization-free routing is the use of hop count, where data is forwarded through nodes that minimize the number of hops to the sink. Another approach is depth-based routing, where nodes closer to the water surface are preferred for forwarding data. Connectivity-based methods also consider the number of neighboring nodes to determine the best forwarding path.

The major advantage of localization-free routing is that it eliminates the need for complex localization mechanisms, thereby reducing energy consumption and computational overhead. These protocols are more suitable for dynamic underwater environments where node positions frequently change.

However, localization-free routing may lead to suboptimal path selection, as decisions are made without precise location information. This can result in increased delay, higher energy consumption in certain nodes, and uneven load distribution across the network.

Despite these limitations, localization-free routing protocols are widely adopted in UWSNs due to their simplicity, practicality, and adaptability to challenging underwater conditions.

3.4 CLUSTERING-BASED ROUTING APPROACHES

Clustering-based routing approaches are widely used in Underwater Wireless Sensor Networks (UWSNs) to improve energy efficiency, scalability, and network lifetime. In this approach, sensor nodes are grouped into clusters, and each cluster is managed by a Cluster Head (CH). The cluster head collects data from member nodes, performs data aggregation, and forwards the processed data to the sink node.

The main objective of clustering is to reduce energy consumption by minimizing long-distance communication. Instead of every node transmitting data directly to the sink, only cluster heads handle long-range communication. This reduces communication overhead and balances energy consumption among nodes.

One of the most important tasks in clustering is cluster head selection. It is generally based on residual energy and can be expressed as:

$$CH = \arg\max (E_{res}) \tag{1}$$

where E_{res}represents the residual (remaining) energy of a node.

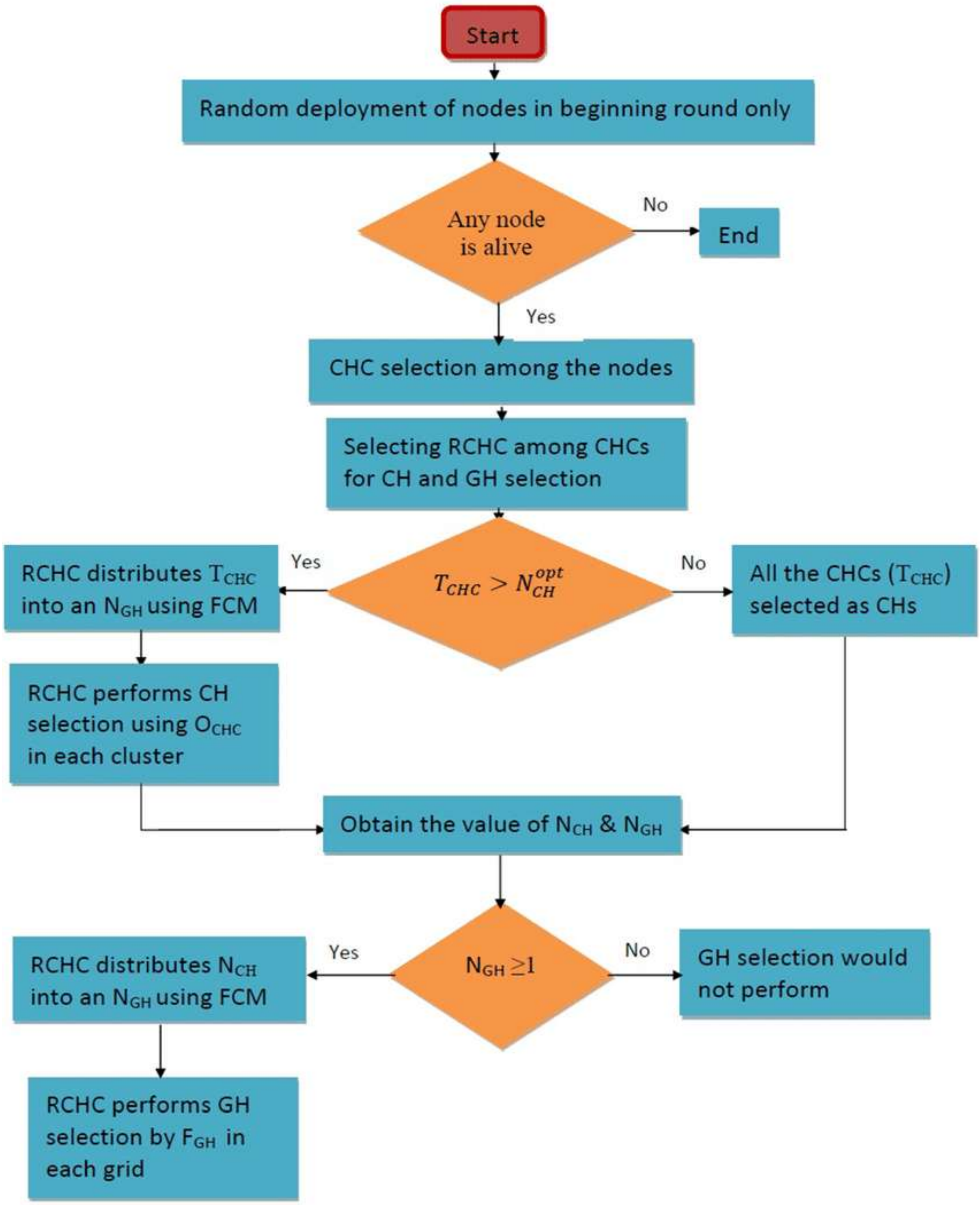

Fig 3.2 Flowchart of cluster head selection process

The clustering process begins by dividing the network into regions. Each node sends a request to become a cluster head. The system evaluates node energies using:

$$E_{max} = \max(E_i) \tag{2}$$

where $E_i$denotes the energy of node i.

The node with the highest energy is selected as the cluster head:

$$CH = N_i \text{where} E_i = E_{max} \tag{3}$$

where $N_i$represents the selected node.

After selection, the cluster head broadcasts its identity, and other nodes join the cluster.

3.4.1 Low Energy Adaptive Clustering Hierarchy (LEACH)

LEACH is one of the earliest clustering-based routing protocols and operates in rounds. It selects cluster heads using a probabilistic threshold function:

$$T(n) = \frac{P}{1-P\left(r\,\mathrm{m}\,\frac{1}{P}\right)} \tag{4}$$

where Prepresents the probability of cluster head selection and $r$denotes the current round.

LEACH consists of two phases: setup phase and steady-state phase. It distributes energy by rotating cluster heads, but it does not consider residual energy, making it less efficient in underwater environments.

3.4.2 Power Efficient Gathering in Sensor Information Systems (PEGASIS)

PEGASIS forms a chain among nodes instead of clusters, where each node communicates with its nearest neighbor. The transmission energy is given by:

$$E_{tx} = k(E_{elec} + \epsilon d^2) \tag{5}$$

where $E_{tx}$is the transmission energy, $k$is the number of data bits, $E_{elec}$is the energy consumed per bit, $d$is the distance between nodes, and $\epsilon$is the amplifier constant.

3.4.3 Energy Efficient Clustering Multi-Hop Routing (EECMR)

EECMR improves clustering by combining energy and distance factors. The selection can be represented as:

$$W = \alpha \frac{E_{res}}{E_{init}} + \beta \frac{d}{L} \tag{6}$$

where $E_{res}$is the residual energy, $E_{init}$is the initial energy, $d$is the distance to the sink, $L$is the network dimension, and $\alpha, \beta$are weighting factors.

EECMR supports multi-hop communication, reducing transmission distance and improving efficiency.

3.4.4 Energy Efficient Routing Based on Layers and Unequal Clusters (EERBLC)

EERBLC forms unequal clusters based on energy and depth. The competition radius is expressed as:

$$R_c \propto \frac{E_{res}}{E_{init}} \tag{7}$$

where $R_c$represents the cluster radius, $E_{res}$is the residual energy, and $E_{init}$is the initial energy.
This approach helps in balancing load among nodes and improving network lifetime.

3.5 DEPTH-BASED ROUTING METHODS

Depth-based routing is widely used in underwater wireless sensor networks because obtaining accurate geographical location information underwater is difficult. These methods use node depth as the main routing parameter, where nodes closer to the water surface are preferred for forwarding data to the sink node.

The major advantage of depth-based routing is that it avoids complex localization techniques, reducing computational overhead and energy consumption. Using pressure sensors, each node determines its depth and forwards data to neighboring nodes with lower depth values, creating an efficient upward data flow from underwater nodes to the sink.

3.5.1 Depth-Based Routing (DBR)

Depth-Based Routing (DBR) is one of the most commonly used routing protocols in UWSNs. It makes routing decisions based solely on the depth information of nodes, without requiring any location or distance calculations.

In DBR, when a node receives a data packet, it compares its depth with the depth value stored in the packet. If the node is closer to the surface than the sender, it forwards the packet; otherwise, it discards it. This ensures that data always moves upward toward the sink node.

The forwarding decision can be simply represented as:

$$Forward \text{ if } d_{current} < d_{previous} \tag{8}$$

where $d_{current}$ is the depth of the current node and $d_{previous}$ is the depth of the transmitting node.

DBR is simple and does not require routing tables or complex computations. It is also robust to topology changes and works well in dynamic underwater environments. However, DBR has some limitations. Nodes near the surface tend to forward more packets, which leads to faster energy depletion. Additionally, multiple nodes may forward the same packet, causing redundancy and increased energy consumption.

3.5.2 Energy Efficient Depth-Based Routing (EEDBR)

Energy Efficient Depth-Based Routing (EEDBR) is an improved version of DBR that addresses its energy-related limitations. In addition to depth, EEDBR considers the residual energy of nodes when making forwarding decisions.

In EEDBR, nodes with higher residual energy are given priority for forwarding packets. This helps in balancing energy consumption across the network and prevents early depletion of specific nodes.

The forwarding decision can be represented as:

$$Priority \propto \frac{E_{res}}{d} \tag{9}$$

where E_{res} represents the residual energy of the node and d represents its depth.

This means that nodes with higher energy and closer proximity to the surface are more likely to forward data. As a result, EEDBR improves network lifetime and energy efficiency compared to DBR.

However, EEDBR introduces additional computational overhead due to energy evaluation and may require more control messages for maintaining node information. Despite these challenges, it provides better performance in terms of energy balance and network stability.

3.6 ENERGY CONSUMPTION AND NETWORK MODEL

Energy consumption is one of the most critical factors in Underwater Wireless Sensor Networks (UWSNs), as sensor nodes are powered by limited battery resources that are difficult to replace or recharge. The performance and lifetime of the network mainly depend on how efficiently energy is utilized during communication.

Unlike terrestrial networks, UWSNs use acoustic communication, which consumes more energy due to high attenuation, noise, and long propagation distances. Therefore, it is essential to model energy consumption accurately to design efficient routing protocols. The energy model in UWSNs mainly includes transmission energy, reception energy, and data aggregation energy.

3.6.1 Acoustic Communication Characteristics

In underwater environments, communication uses acoustic signals instead of radio waves because electromagnetic waves suffer severe attenuation in water. Acoustic communication has low bandwidth, high propagation delay, and high energy consumption. Since sound travels in water at about 1500 m/s, delays are much higher than radio communication in air. Signal quality is also affected by multipath fading, noise, and absorption losses.

The signal attenuation in underwater communication can be expressed as:

$$A(d,f) = d^k \cdot a(f)^d \tag{10}$$

where $A(d,f)$represents attenuation, $d$is the distance between transmitter and receiver, $f$is the signal frequency, $k$is the spreading factor, and $a(f)$is the absorption coefficient.

This shows that attenuation increases with both distance and frequency, making long-distance communication more energy-consuming.

3.6.2 Channel Attenuation and Noise Model

The underwater channel is affected by various types of noise that degrade communication performance. The major noise sources include turbulence noise, shipping noise, wave noise, and thermal noise.

The overall noise can be represented as a combination of different components:

$$N(f) = N_t(f) + N_s(f) + N_w(f) + N_{th}(f) \tag{11}$$

where $N(f)$ is the total noise, $N_t(f)$ is turbulence noise, $N_s(f)$ is shipping noise, $N_w(f)$ is wave noise, and $N_{th}(f)$ is thermal noise.

These noise components depend on environmental conditions and frequency, and they affect the signal-to-noise ratio, leading to communication errors and increased energy consumption due to retransmissions.

3.6.3 Transmission and Reception Energy Model

Energy consumption during communication mainly occurs during transmission and reception of data. The energy required to transmit data depends on both the size of the data and the distance between nodes.

The transmission energy can be expressed as:

$$E_{tx}(k,d) = k \cdot (E_{elec} + E_{amp}) \tag{12}$$

where E_{tx} is the transmission energy, k is the number of bits, E_{elec} is the energy required for electronic circuits, and E_{amp} is the energy required for signal amplification.

The amplification energy depends on distance and can be written as:

$$E_{amp} \propto d^2 \tag{13}$$

where d is the distance between nodes.

The reception energy is given by:

$$E_{rx}(k) = k \cdot E_{elec} \tag{14}$$

where E_{rx} is the energy required to receive data.

In addition to transmission and reception, data aggregation also consumes energy, which can be expressed as:

$$E_{agg} = k \cdot E_{DA} \tag{15}$$

where E_{agg} is aggregation energy and E_{DA} is the energy required for processing one bit of data.

The total energy consumption in the network can be represented as:

$$E_{total} = E_{tx} + E_{rx} + E_{agg} \tag{16}$$

where E_{total} represents the total energy consumed during communication.

3.7 ROUTING STRATEGIES

Routing strategies in Underwater Wireless Sensor Networks (UWSNs) define how data is transmitted from sensor nodes to the sink node efficiently. Due to challenges such as high energy consumption, long propagation delay, and dynamic topology, selecting an appropriate routing strategy is essential for improving network performance and lifetime.

Different routing strategies are adopted in UWSNs based on network size, node deployment, and energy constraints. The most common strategies include single-hop communication, multi-hop communication, cluster-based transmission, and relay-based communication. Each strategy has its own advantages and limitations depending on the application requirements.

3.7.1 Single-Hop and Multi-Hop Communication

In single-hop communication, each sensor node directly transmits its data to the sink node. This approach is simple and requires less coordination among nodes. However, it consumes a large amount of energy, especially when the distance between the node and the sink is high.

The transmission energy can be represented as:

$$E_{tx} \propto d^2 \tag{17}$$

where $d$represents the distance between the node and the sink. This shows that energy consumption increases significantly with distance.

In multi-hop communication, data is forwarded through intermediate nodes in shorter steps before reaching the sink, reducing energy consumption and extending network lifetime. It is suitable for large-scale UWSNs as it balances energy usage, though it may introduce additional delay due to multiple transmissions.

3.7.2 Cluster-Based Data Transmission

In cluster-based data transmission, sensor nodes are grouped into clusters, and each cluster has a cluster head. Member nodes send their data to the cluster head, which performs data aggregation and forwards the processed data to the sink node.

This approach reduces redundant transmissions and minimizes energy consumption by limiting long-distance communication to cluster heads only. Data aggregation helps in eliminating duplicate information, further improving efficiency.

The total energy consumption in cluster-based transmission can be expressed as:

$$E_{total} = E_{tx} + E_{rx} + E_{agg} \quad (18)$$

where $E_{tx}$is transmission energy, $E_{rx}$is reception energy, and $E_{agg}$is data aggregation energy.

Cluster-based transmission improves scalability and is widely used in UWSNs. However, the performance depends heavily on efficient cluster head selection and load balancing.

3.7.3 Relay-Based Communication

Relay-based communication is a routing strategy in which intermediate nodes, known as relay nodes, assist in forwarding data between source nodes and the sink. This approach is especially useful when direct communication is not feasible due to long distances or obstacles.

In this method, relay nodes are selected based on parameters such as residual energy, distance, and connectivity. Data is transmitted from the source node to the relay node and then forwarded to the next node or sink.

The selection of relay nodes can be represented as:

$$Relay = \arg\max\,(E_{res}) \quad (19)$$

where E_{res}represents the residual energy of the node.

Relay-based communication reduces transmission distance and improves energy efficiency. It also enhances reliability by providing alternative paths for data transmission. However, improper selection of relay nodes may lead to uneven energy consumption and increased delay.

3.8 CLUSTER HEAD AND RELAY NODE SELECTION

Cluster head and relay node selection are crucial in underwater wireless sensor networks, as they directly influence energy consumption, transmission efficiency, and network lifetime. Cluster heads collect and aggregate data, while relay nodes forward it to the sink node. Selecting suitable nodes based on factors such as probability, energy, and depth helps balance energy usage, reduce communication overhead, and improve overall network performance.

3.8.1 Probabilistic Selection Methods

In probabilistic selection methods, nodes are selected as cluster heads based on a predefined probability. Each node independently decides whether to become a cluster head using a threshold function.

The selection process can be represented as:

$$CH = \begin{cases} 1, & \text{if } rand < T(n) \\ 0, & \text{otherwise} \end{cases}$$

where $rand$ is a random number between 0 and 1, and $T(n)$is the threshold value.

This method is simple and helps in distributing the cluster head role among nodes over time. It is commonly used in protocols like LEACH. However, probabilistic selection does not consider important parameters such as residual energy or node position. As a result, nodes with low energy may be selected as cluster heads, leading to faster energy depletion and reduced network lifetime.

3.8.2 Energy-Based Selection

Energy-based selection methods choose cluster heads and relay nodes based on the residual energy of nodes. Nodes with higher remaining energy are given priority, ensuring efficient utilization of network resources.

The selection can be expressed as:

$$CH = \arg\max(E_{res}) \tag{21}$$

where E_{res} represents the residual energy of the node.

Similarly, relay nodes can also be selected based on maximum energy:

$$Relay = \arg\ \max\ (E_{res}) \tag{22}$$

This approach balances energy consumption, prevents early node failure, and improves network lifetime and reliability compared to probabilistic methods. However, energy-based selection may ignore factors such as distance and communication delay, which can affect overall performance.

3.8.3 Depth-Based Selection

Depth-based selection methods use the depth of nodes as the main parameter for selecting cluster heads or relay nodes. In underwater environments, nodes closer to the surface are preferred because they are nearer to the sink node.

The selection condition can be expressed as:

$$Select\ node \text{ if } d_{current} < d_{threshold} \quad (23)$$

where $d_{current}$ represents the depth of the node and $d_{threshold}$ is a predefined depth limit.

In relay selection, nodes with lower depth values are preferred for forwarding data, as they help in moving data upward toward the sink. Depth-based selection is simple and does not require complex computations or localization techniques. However, it may lead to uneven energy consumption, as nodes near the surface are used more frequently and may deplete their energy faster.

3.9 EXISTING METHOD RESULTS

Table 3.1 Simulation Parameters Used for Performance Evaluation

Parameters	Value
Network Area	500 mx500 mx500 m
Number of Nodes	100–300
Data packet size	200bits
Transmission Distance	150 m
Transmission Frequency	30 kHz.
Initial Energy Receiver power constant(P_0) Aggregation energy constant (E_{DA0}) Bit transmission energy constant	5 Joules 5×10^{-10} 5×10^{-9} 5×10^{-9}

The simulation and network setup parameters used for evaluating the proposed Gravitational Search Algorithm (GSA)-based cluster routing protocol in the Underwater Wireless Sensor Network (UWSN) are presented in Table 2. The underwater environment is considered as a three-dimensional network area of 500 m × 500 m × 500 m, in which 100 to 300 sensor nodes are randomly deployed. Each sensor node is initialized with 5 Joules of energy, and acoustic communication is performed at a transmission frequency of 30 kHz. The packet size is fixed at 200 bits, while the maximum transmission range is set to 150 m. In addition, important energy-related parameters such as receiver power constant (P_0), aggregation energy constant (EDA_0), and bit transmission energy constant are considered for estimating transmission, reception, and data aggregation energy. These simulation parameters are used to analyze the effectiveness of the proposed GSA-based routing protocol in terms of network lifetime, residual energy, throughput, and dead node performance. Performance metrics based on the expectancy of algorithms.

Table 3.2 Comparative Analysis of Routing Algorithms Based on Network Lifetime

Algorithms	Number of rounds	First node death	Last node death	Rate of exploitation
Proposed- GSA	1367	1149	1367	Slope Linear
EECMR	915	847	915	Exponential
EERBLC	1206	613	1206	Linear
LEACH	587	221	587	Intermediate

The performance comparison of the proposed GSA-based routing protocol with EECMR, EERBLC, and LEACH using metrics such as number of rounds, first node death, last node death, and rate of exploitation is shown in Table 2. The proposed GSA achieved the highest network lifetime with 1367 rounds, compared to EECMR (915), EERBLC (1206), and LEACH (587). It also delayed the first node death to 1149 rounds, indicating better stability and energy balancing than the existing methods. The last node death for GSA occurred at 1367 rounds, showing prolonged network operation. In terms of exploitation rate, the proposed GSA followed a linear trend, whereas EECMR showed exponential behavior and LEACH showed intermediate performance. Overall, the proposed GSA-based routing protocol demonstrated better lifetime, stability, and energy efficiency compared to the existing algorithms.

CHAPTER 4

PROPOSED METHOD

CHAPTER 4
PROPOSED METHOD

4.1 INTRODUCTION

The proposed methodology improves energy efficiency, network lifetime, and data transmission reliability in underwater wireless sensor networks by combining data reduction, optimized clustering, and intelligent routing. It addresses key underwater challenges such as high energy consumption, redundant transmissions, limited bandwidth, attenuation, and long propagation delays.

Underwater sensor nodes are randomly deployed in a three-dimensional environment to monitor parameters such as temperature, pressure, and salinity. Since raw sensed data may contain correlated and redundant information, a correlation–entropy-based feature selection method is first applied to retain only the most relevant data and remove unnecessary features.

After feature selection, a lightweight differential compression technique is used to reduce transmission size. Instead of sending complete readings every time, nodes transmit only the difference between current and previous values, lowering packet size and communication cost with minimal overhead.

To improve communication efficiency, the network is organized into clusters using the Gravitational Search Algorithm. Sensor nodes are evaluated based on residual energy, node density, and distance to the sink node, and nodes with the best fitness values are selected as Cluster Heads for balanced energy usage and efficient aggregation.

After clustering, a Deep Reinforcement Learning-based routing mechanism is applied for adaptive and energy-aware data forwarding. The DRL agent learns optimal routes using factors such as residual energy, transmission distance, link quality, and packet delivery performance, enabling dynamic relay node and path selection.

The communication follows a multi-hop strategy where ordinary nodes send compressed data to cluster heads, and cluster heads forward aggregated data to the sink through selected relay nodes. By integrating feature selection, compression, GSA-based clustering, and DRL-based routing, the proposed system significantly enhances network performance, stability, and lifetime.

4.2 SENSOR NODE DEPLOYMENT

The sensor node deployment in the proposed system is carried out in a three-dimensional underwater environment to simulate realistic ocean monitoring conditions. The network consists of a large number of sensor nodes randomly distributed at different depths to collect environmental data such as temperature, pressure, and salinity.

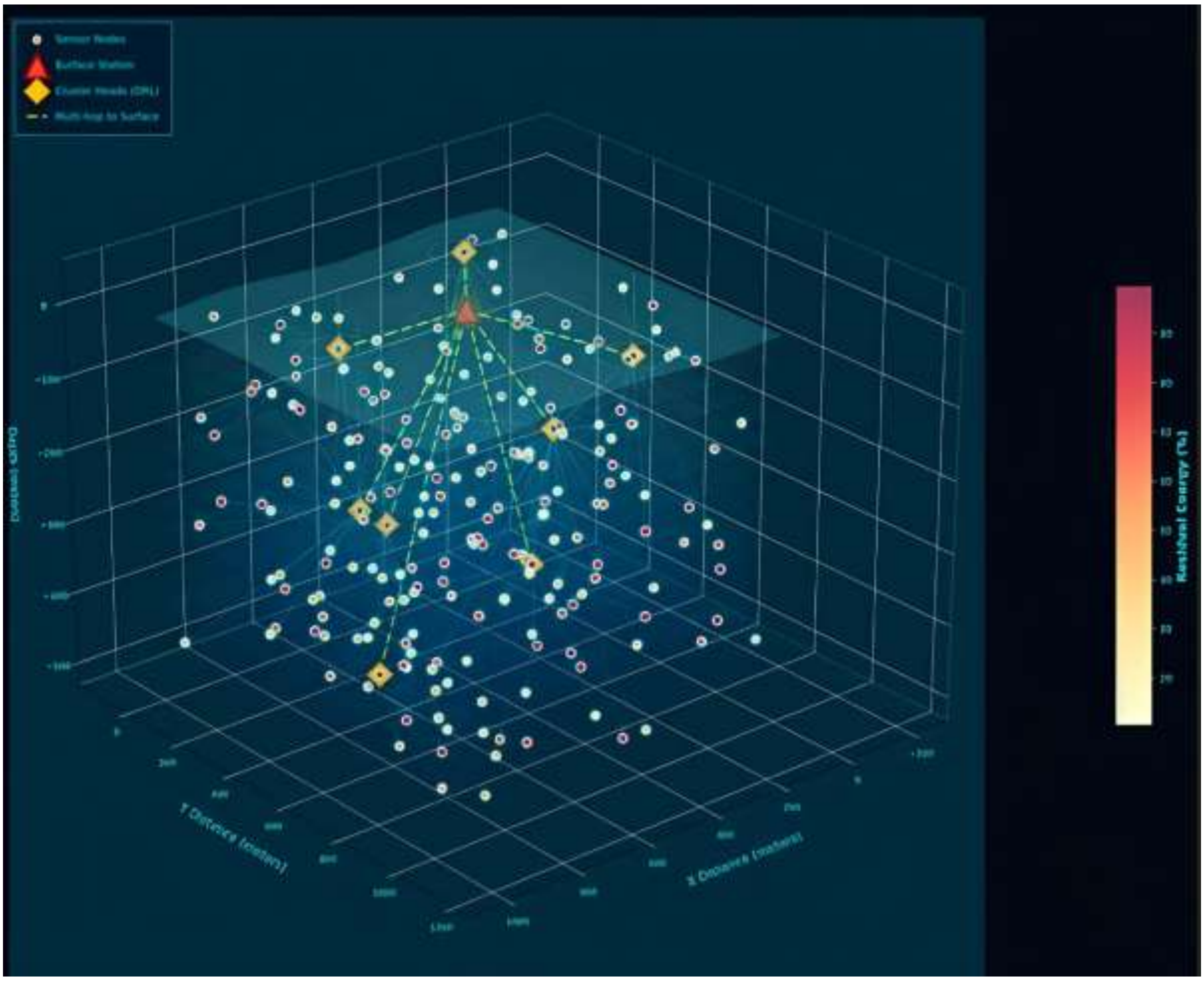

Fig 4.1 3D Sensor Node Deployment in UWSN

As shown in the figure, the deployment area is modeled as a 3D space with defined X, Y, and depth (Z) coordinates. Sensor nodes are represented as distributed points within this space, while the sink node (surface station) is positioned at the water surface to collect data from the network. Cluster heads are dynamically selected nodes that act as intermediate aggregators and are responsible for forwarding data toward the sink.

The deployment follows a random distribution strategy to reflect real-world underwater conditions where precise placement of nodes is difficult. Each node is equipped with sensing, processing, and acoustic communication capabilities. Nodes communicate with neighboring nodes using multi-hop communication due to the high attenuation of acoustic signals over long distances.

The figure also illustrates the multi-hop communication paths between sensor nodes, cluster heads, and the sink node. Data is first transmitted from sensor nodes to cluster heads and then forwarded through intermediate relay nodes to the surface station. This hierarchical communication structure reduces long-distance transmissions and improves energy efficiency.

Additionally, the color variation of nodes represents different residual energy levels, indicating the dynamic energy distribution across the network. Nodes with higher energy are more likely to be selected as cluster heads or relay nodes, ensuring balanced energy consumption.

This deployment model provides a realistic foundation for implementing the proposed preprocessing, clustering, and DRL-based routing techniques, enabling efficient data transmission and extended network lifetime.

4.3 DATA PREPROCESSING

Data preprocessing is an important step in the proposed system to reduce redundant transmissions and improve energy efficiency in underwater wireless sensor networks. In underwater environments, nearby sensor nodes often generate similar data, and transmitting all of it increases communication overhead and battery depletion.

To address this, the system applies preprocessing before clustering and routing using two techniques: correlation–entropy-based feature selection and lightweight differential compression. Raw sensor data is processed to remove redundancy and reduce data size, ensuring that only relevant and compact information is transmitted, which lowers transmission cost and improves overall network performance.

4.3.1 Correlation–Entropy Feature Selection

Correlation–entropy feature selection is a hybrid data reduction technique used in the proposed system to eliminate redundant and less informative sensor data before transmission in Underwater Wireless Sensor Networks (UWSNs). Due to dense deployment of sensor nodes in underwater environments, neighboring nodes often observe similar physical conditions such as temperature, pressure, and salinity. This results in highly correlated data, where multiple nodes transmit nearly identical information, leading to unnecessary energy consumption and increased communication overhead.

To address this issue, correlation analysis is first applied to identify the similarity between sensor readings. The correlation between two features X and Y is calculated as:

$$Corr(X,Y) = \frac{Cov(X,Y)}{\sigma_X \sigma_Y} \tag{24}$$

where $Cov(X,Y)$ represents covariance between features, and σ_X, σ_Y denote their standard deviations. A correlation value close to 1 indicates strong similarity, while a value close to 0 indicates independence.

If the correlation exceeds a predefined threshold θ, i.e.

$$Corr(X,Y) > \theta$$

one of the features is considered redundant and removed from the dataset.

This step ensures that duplicate or highly similar information is not transmitted, thereby reducing data redundancy.

After correlation filtering, entropy-based analysis is applied to evaluate the significance of the remaining features. Entropy, denoted as $H(X)$, measures the amount of information or uncertainty present in a feature and is given by:

$$H(X) = -\sum p(x) \log\, p(x) \tag{25}$$

where $p(x)$ represents the probability distribution of the data values. A higher entropy value indicates that the feature contains more variability and useful information, while lower entropy indicates predictable or less informative data.

In the proposed approach, features are selected only if they satisfy the condition:

$$H(X) > \theta_e \tag{26}$$

where θ_e is the entropy threshold. This ensures that only highly informative features are retained.

The combined condition for feature selection can be summarized as:

$$Corr(X,Y) < \theta \text{and} H(X) > \theta_e \tag{27}$$

This dual filtering mechanism ensures the selected data is non-redundant and information-rich, significantly reducing transmitted data, energy consumption, and improving network efficiency. It also enhances data quality by removing noise and irrelevant information, making clustering and routing more effective.

4.3.2 Lightweight Differential Compression

After selecting the most relevant features, lightweight differential compression is applied to further reduce the size of transmitted data. This technique is particularly suitable for underwater environments where sensor readings change gradually over time, resulting in minimal differences between consecutive values.

Instead of transmitting complete sensor readings, each node computes the difference between the current reading $D_{current}$and the previous reading $D_{previous}$

$$\Delta D = D_{current} - D_{previous} \tag{28}$$

where ΔD represents the change in sensor data. Since environmental variations are typically small, ΔD values are significantly smaller than the original data values, allowing them to be encoded using fewer bits.

This reduction in data size directly impacts energy consumption, as transmission energy is proportional to the number of bits transmitted. Therefore, smaller packet sizes lead to lower energy usage and improved network lifetime.

To further optimize communication, a threshold-based transmission mechanism is implemented. A predefined threshold $\delta$is used to determine whether the change in data is significant enough to transmit:

$$| \Delta D | > \delta \tag{29}$$

If the absolute difference exceeds the threshold, the data is transmitted; otherwise, transmission is skipped. This prevents unnecessary communication of minor variations, which do not significantly affect the overall monitoring process.

Additionally, the compression efficiency can be understood in terms of data reduction:

$$Compression\ Ratio = \frac{Size_{original}}{Size_{compressed}} \tag{30}$$

A higher compression ratio indicates better reduction in data size and improved efficiency.

The proposed compression method is lightweight, meaning it requires minimal computational complexity and memory resources. This makes it highly suitable for resource-constrained underwater sensor nodes. It avoids complex encoding schemes, thereby reducing processing delay and energy consumption. By combining differential encoding, threshold-based filtering, and efficient data representation, the proposed compression technique significantly reduces communication overhead, improves bandwidth utilization, and enhances overall network performance.

4.4 GSA-OPTIMIZED CLUSTER FORMATION

Clustering is essential for improving energy efficiency and scalability in underwater wireless sensor networks. In the proposed system, cluster formation is optimized using the Gravitational Search Algorithm, inspired by Newton's law of gravity, to select optimal cluster heads and create balanced clusters.

Unlike traditional static or probabilistic methods, GSA dynamically selects cluster heads based on residual energy, node density, distance to the sink, and communication cost. Each sensor node is treated as an agent, where nodes with higher fitness values gain stronger gravitational mass and are more likely to be chosen as cluster heads.

The fitness of each node is calculated using a multi-parameter function:

$$F_i = \alpha E_{res} + \beta \frac{1}{d_i} + \gamma D_i + \delta C_i \tag{31}$$

where $F_i$represents the fitness of node i, $E_{res}$is the residual energy, $d_i$is the distance to the sink, $D_i$represents node density, and $C_i$represents communication cost. The coefficients $\alpha, \beta, \gamma,$and $\delta$are weighting factors that control the importance of each parameter.

Nodes with higher fitness values obtain higher mass and are more likely to attract other nodes, leading to optimal cluster head selection.

4.4.1 Cluster Head Selection using GSA

Cluster head selection is a critical task in clustering-based routing, as cluster heads handle data aggregation, coordination, and communication with the sink node. In underwater environments, poor selection can cause rapid energy depletion, load imbalance, and reduced network lifetime.

To address this, the proposed system uses the Gravitational Search Algorithm for optimal cluster head selection. Each sensor node is treated as an agent, and its suitability as a cluster head is evaluated using a fitness function.

The fitness of each node is calculated based on multiple parameters including residual energy, distance to the sink, node density, and communication cost. This can be expressed as:

$$F_i = \alpha E_{res} + \beta \frac{1}{d_i} + \gamma D_i + \delta C_i \tag{32}$$

where F_i represents the fitness of node i, E_{res} denotes residual energy, d_i is the distance to the sink, D_i represents node density, and C_i denotes communication cost. The coefficients $\alpha, \beta, \gamma,$ and δ are weighting factors that balance the contribution of each parameter.

Nodes with higher residual energy and favorable network conditions obtain higher fitness values, making them strong candidates for cluster head selection.
After computing fitness values, the mass of each node is determined as a normalized value:

$$M_i = \frac{F_i - F_{min}}{F_{max} - F_{min}} \tag{33}$$

where M_i represents the mass of node i, and F_{max}, F_{min} denote the maximum and minimum fitness values in the network.

In GSA, nodes interact with each other through gravitational forces. Nodes with higher mass exert stronger attraction, influencing other nodes in the search space. The gravitational force between nodes is given by:

$$F_{ij} = G \cdot \frac{M_i \cdot M_j}{R_{ij} + \epsilon} \tag{34}$$

where G is the gravitational constant, R_{ij} is the distance between nodes i and j, and ϵ is a small constant to avoid instability.

The total force acting on each node is used to compute acceleration, which guides the movement of nodes toward optimal solutions. Over multiple iterations, nodes converge toward positions representing optimal cluster head candidates.

Additionally, a time-varying gravitational constant $G(t)$is used to control the convergence speed:

$$G(t) = G_0 \cdot e^{-\lambda t} \tag{35}$$

where $G_0$is the initial gravitational constant, $\lambda$is a decay factor, and $t$represents the iteration number.

Finally, nodes with the highest fitness and mass are selected as cluster heads. This ensures that nodes with sufficient energy and better communication conditions handle the clustering process, leading to balanced energy usage and improved network performance.

4.4.2 Cluster Formation Process

Once optimal cluster heads are selected using GSA, the next step is to organize the network into clusters. The cluster formation process aims to group sensor nodes efficiently such that communication energy is minimized and network load is evenly distributed.

Each sensor node determines its association with a cluster head based on distance, signal strength, and communication cost. The Euclidean distance between a node and a cluster head is calculated as:

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2} \tag{36}$$

where $(x_i, y_i, z_i)$and (x_j, y_j, z_j)represent the coordinates of the sensor node and cluster head, respectively.

Each node selects the cluster head that minimizes the communication distance:

$$CH = \arg\ \min\ (d_{ij}) \tag{37}$$

This ensures that nodes join the nearest cluster head, reducing transmission energy and improving communication efficiency.

A load balancing mechanism limits the number of member nodes per cluster head, assigning extra nodes to nearby cluster heads to ensure uniform cluster distribution.

After cluster formation, communication occurs in two stages:

- Intra-cluster communication, where member nodes transmit data to the cluster head using single-hop transmission
- Inter-cluster communication, where cluster heads forward aggregated data to the sink node using multi-hop routing

4.5 DRL-BASED ENERGY-AWARE ROUTING

Routing in underwater wireless sensor networks is challenging due to varying node energy, unstable links, high propagation delay, and dynamic topology. Traditional fixed-rule routing methods are not effective in such changing environments.

To overcome this, the proposed system uses Deep Reinforcement Learning for energy-aware routing. The DRL agent learns optimal routing decisions by interacting with the network, dynamically selecting the best forwarding nodes based on current conditions to maximize energy efficiency, packet delivery ratio, and network lifetime.

4.5.1 State, Action, and Reward Definition

In the proposed Deep Reinforcement Learning-based routing framework, routing is modeled as a learning problem where the agent interacts with the network using three key components: state, action, and reward. These elements help the agent learn and improve routing decisions over time.

The state represents the current network condition, including residual node energy, distance to the sink, link quality, compressed packet size, and node connectivity. This information allows the agent to evaluate routing options effectively, preferring nodes with higher energy and more stable communication links. The state representation can be expressed as:

$$S_t = E_{res}, d, LQ, P_{size} \tag{38}$$

where E_{res}denotes residual energy, $d$represents transmission distance, LQindicates link quality, and P_{size}represents packet size after compression.

The action corresponds to selecting the next-hop node from a set of neighboring nodes. At each step, the DRL agent evaluates all available neighbor nodes and selects the one that is expected to provide the best performance in terms of energy efficiency and communication reliability. The action space consists of all possible forwarding nodes within the communication range.

The reward function provides feedback to the agent based on the quality of the selected action. It plays a crucial role in guiding the learning process. In the proposed system, the reward is designed to encourage energy-efficient and reliable routing while penalizing poor decisions. A simplified reward formulation is given by:

$$R = \alpha E_{res} + \beta \frac{1}{d} + \gamma LQ \tag{39}$$

where α, β,and $\gamma$are weighting factors.

A higher reward is obtained when the selected node has higher energy, shorter distance, and better link quality. Conversely, a lower reward is assigned when the routing decision leads to higher energy consumption or unreliable communication.

Over time, the DRL agent learns the relationship between state and reward. It updates its policy to maximize long-term cumulative rewards, which results in improved routing decisions. This learning capability allows the system to adapt to changing network conditions and maintain optimal performance.

4.5.2 Relay Node Selection using DRL

Relay node selection is a key component of the routing process, as it determines how efficiently data is transmitted through the network. In the proposed system, DRL is used to dynamically select relay nodes based on current network conditions instead of relying on static or predefined criteria.

At each step, the DRL agent observes the state of neighboring nodes and evaluates their suitability as relay nodes. The selection is based on multiple parameters such as residual energy, distance to the sink, link quality, and communication cost. By considering multiple factors simultaneously, the system ensures that the selected node provides a balanced trade-off between energy efficiency and communication reliability.

The optimal relay node is selected using:

$$a^* = \arg\max Q(s, a) \tag{40}$$

where $Q(s, a)$represents the expected reward of selecting action $a$in state s.

During the initial stages, the DRL agent explores different routing options by selecting various nodes. This helps the system learn the behavior of the network under different conditions. As the learning process progresses, the agent gradually shifts toward selecting the best-performing nodes based on learned experience.

This exploration–exploitation balance allows the system to continuously improve its routing decisions. If a node's energy decreases or its link quality becomes poor, the DRL agent automatically reduces its selection probability and chooses a better alternative. This prevents overuse of certain nodes and ensures balanced energy consumption across the network.

4.5.3 Adaptive Multi-Hop Routing Mechanism

The proposed routing mechanism follows an adaptive multi-hop strategy to minimize energy consumption and improve data delivery efficiency. In underwater environments, direct communication between nodes and the sink requires high transmission power, which leads to rapid energy depletion. Multi-hop routing overcomes this issue by dividing the communication into shorter transmissions.

In this approach, data transmission occurs in stages. Sensor nodes first send data to cluster heads, and cluster heads forward the data through relay nodes selected by the DRL agent. At each hop, the next node is selected dynamically based on current network conditions.

The routing decision is guided by a cost function that considers energy and communication quality:

$$Cost = \frac{d}{E_{res}} + \frac{1}{LQ} \tag{41}$$

where $d$represents distance, $E_{res}$is residual energy, and $LQ$is link quality.

Nodes with lower cost are preferred, as they provide better energy efficiency and communication reliability. The DRL agent uses this information indirectly through the reward mechanism to learn optimal routing paths.

4.6 WORKING OF THE PROPOSED SYSTEM

The overall working of the proposed system is designed as a structured and sequential process that integrates data preprocessing, optimization-based clustering, and intelligent routing to achieve efficient communication in Underwater Wireless Sensor Networks (UWSNs). The system aims to minimize energy consumption, reduce redundant data transmission, and improve network lifetime while maintaining reliable communication in dynamic underwater environments.

The process begins with the deployment of sensor nodes in a three-dimensional underwater region. These sensor nodes are randomly distributed at different depths to simulate realistic ocean monitoring scenarios. Each node is capable of sensing environmental parameters such as temperature, pressure, salinity, and pollution levels. The collected data is then prepared for transmission toward the sink node located at the water surface.

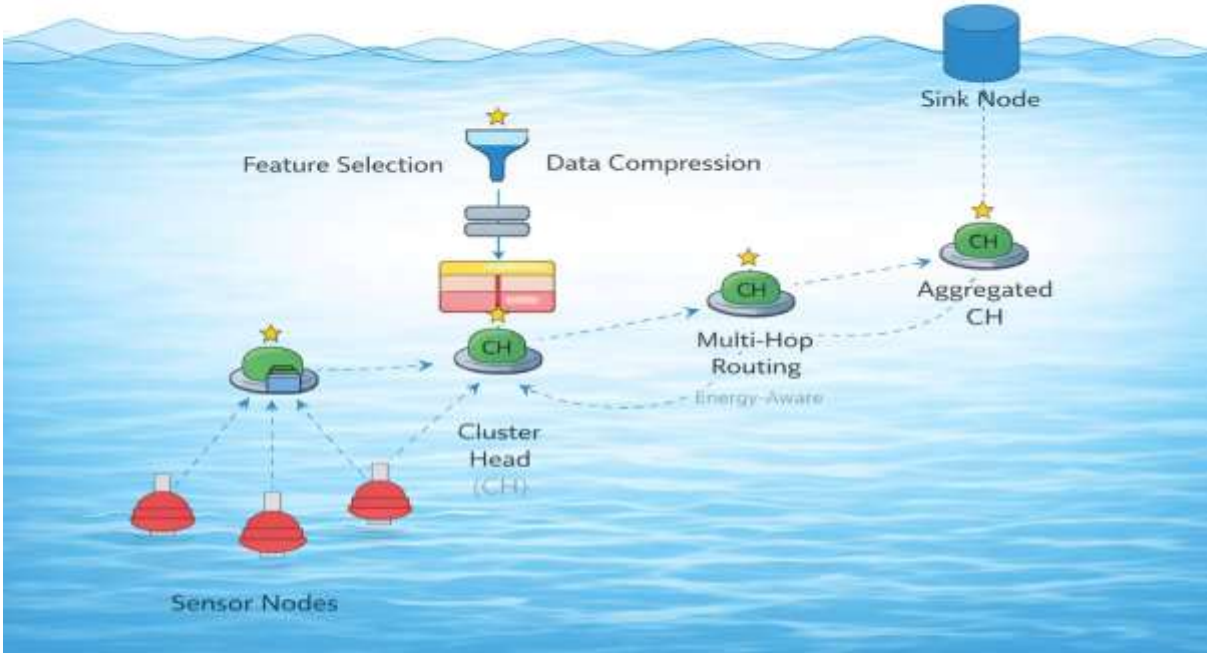

Fig 4.2 Architecture of DRL-Enhanced GSA-Based Compression-Aware Routing in UWSN

As shown in the figure, the system follows a multi-stage pipeline to improve efficiency and reduce unnecessary communication. After data collection, the preprocessing stage applies correlation–entropy-based feature selection to remove redundant and less informative data. Correlation identifies duplicate readings from neighboring nodes, while entropy retains only meaningful and high-information content data.

Following feature selection, lightweight differential compression is applied to reduce the size of the data. Instead of transmitting complete sensor readings, only the difference between consecutive readings is transmitted. This significantly reduces the number of bits transmitted and lowers the energy required for communication. The combination of feature selection and compression ensures that only essential and compact data is forwarded to the next stage.

After preprocessing, the system performs clustering using the GSA-based optimization approach. In this stage, cluster heads (CH) are selected based on parameters such as residual energy, distance to the sink, and node density. The selected cluster heads act as local coordinators that collect data from nearby sensor nodes. As shown in the figure, multiple sensor nodes are connected to a cluster head, forming a cluster. This hierarchical structure reduces direct communication between sensor nodes and the sink, thereby conserving energy.

Once clusters are formed, cluster heads perform data aggregation to combine data from multiple nodes into a single packet. This aggregation further reduces redundancy and minimizes the amount of data that needs to be transmitted through the network. The aggregated data is then forwarded toward the sink node using an energy-aware routing mechanism.

The routing stage is implemented using a DRL-based approach, which enables intelligent and adaptive routing decisions. Instead of following fixed paths, the DRL agent dynamically selects the best relay nodes for forwarding data. As shown in the figure, data is transmitted through multiple intermediate cluster heads or relay nodes using a multi-hop communication strategy. This approach reduces long-distance transmission and distributes energy consumption across the network.

At each hop, the DRL agent evaluates the network conditions, including node energy levels, link quality, and transmission distance. Based on this information, it selects the most suitable next-hop node. If a node has low energy or poor connectivity, it is avoided, and an alternative path is selected. This ensures reliable and efficient data transmission even in dynamic underwater conditions.

Another important aspect of the proposed system is that it is compression-aware. Since the data has already been compressed in the preprocessing stage, the routing process considers the size of the data packets. Nodes that can efficiently handle larger packets are selected carefully to avoid congestion and packet loss.

The final stage of the process is data delivery to the sink node. The sink node collects data from multiple cluster heads and relay nodes and processes it for further analysis. Since the data has already been filtered, compressed, and aggregated, the sink receives only meaningful and optimized information.

The entire system operates in a continuous cycle, where sensing, preprocessing, clustering, and routing are repeated over time. As the network operates, the DRL agent continuously learns and updates its routing strategy based on feedback from the environment. This learning capability allows the system to adapt to changes such as energy depletion, node failures, and variations in communication quality.

Overall, the proposed system provides an efficient and intelligent solution for underwater communication. By combining data reduction techniques, optimization-based clustering, and DRL-based adaptive routing, the system significantly reduces energy consumption, improves data delivery performance, and extends the overall network lifetime.

4.7 IMPLEMENTATION OVERVIEW (ENHANCED WITH CODE DETAILS)

4.7.1 Energy-Efficient Routing Algorithm for UWSNs

Input:

Si: Source Node

CH: Cluster Head

St: Sink Node

Eres: Residual Energy

D (i, j): Distance between nodes

P_{size} : Packet size after compression

Algorithm Steps:

1. Initialize network with N sensor nodes having random 3D coordinates and energy values
2. Select cluster heads based on highest residual energy values
3. For each node Si:
    - Generate sensor data using data engine
    - Apply feature selection and differential compression
4. Compute routing cost for each candidate cluster head
5. Select optimal next-hop node using DRL-based Q-value estimation
6. Transmit data from node to cluster head

7. Forward aggregated data from cluster head to sink
8. Update energy and repeat until simulation ends

Output:
Optimized energy-efficient routing path

4.7.2 Overview of Code Structure

The implementation is developed in Python using object-oriented programming concepts and is divided into modules such as node deployment, data preprocessing, clustering, DRL-based routing, and performance evaluation. It uses NumPy for numerical operations, PyTorch for building the Deep Q-Network, and Matplotlib for visualization. Each module is implemented as classes or functions to ensure modularity and scalability.

The system follows a layered workflow where each module processes data and passes output to the next stage. Custom classes handle tasks such as data compression, state representation, action selection, and policy updates, improving code reusability and maintainability.

The Deep Q-Network uses a multi-layer fully connected neural network that maps input states to Q-values for routing decisions. Training is performed using gradient-based optimization in PyTorch, while a simulation loop over multiple episodes allows the DRL agent to learn from state transitions, rewards, and actions.

The framework also integrates performance evaluation and visualization. Metrics such as energy consumption and routing efficiency are measured during simulation, while graphical and 3D visualizations help analyze node deployment, cluster formation, and routing paths. Overall, the code is modular, extensible, and efficient for integrating preprocessing, clustering, and DRL-based routing into one system.

4.7.3 Network Initialization and Deployment

The network initialization phase creates a simulated underwater environment where sensor nodes are deployed randomly in a 3D space. Each node is assigned coordinates and energy values, representing real-world underwater conditions.

The positions of nodes are generated using random distributions, ensuring uniform coverage of the deployment area. The sink node is placed at a fixed position above the water surface to collect data.

Each node maintains attributes such as:
Position (x, y, z coordinates)
Residual energy
Communication range

Distance between nodes is calculated dynamically during routing, enabling accurate modeling of communication paths. This setup forms the foundation for all subsequent operations in the system.

In addition to basic initialization, the deployment process also simulates realistic underwater characteristics such as varying node depths and irregular spatial distribution. The z-coordinate of each node represents depth below the water surface, which plays an important role in communication and routing decisions.

The residual energy of each node is initialized within a specific range to reflect heterogeneous energy conditions. This variation allows the system to evaluate energy-aware algorithms effectively, particularly during cluster head selection and routing.

The communication range of nodes is implicitly considered during neighbor selection, where nodes can only communicate with other nodes within a certain distance. This constraint ensures realistic modeling of acoustic communication in underwater environments, were signal attenuation limits long-distance transmission.

Furthermore, the deployment module supports dynamic interaction between nodes by continuously updating node states during simulation. As nodes transmit data, their energy levels decrease, and this affects future routing and clustering decisions. This dynamic behavior closely mimics real-world underwater sensor networks.

The network initialization also prepares essential data structures such as node position arrays and energy vectors, which are used throughout the simulation for distance calculations, routing decisions, and visualization. These structures enable efficient computation and fast access to node information.

Overall, the network initialization and deployment stage provides a realistic and flexible simulation environment, ensuring that the proposed algorithms are tested under conditions that closely resemble actual underwater scenarios.

4.7.4 Data Preprocessing Module

The preprocessing module is responsible for optimizing the data before transmission. It reduces redundancy and ensures efficient use of network resources. This module is implemented using a data engine associated with each node. It generates sensor readings and applies feature selection to retain only relevant information.

Differential compression is applied to reduce packet size by transmitting only the change in data values. Since environmental parameters change gradually, this approach significantly reduces the number of transmitted bits.

Additionally, the module calculates compression ratio, which helps in evaluating the efficiency of data reduction. By minimizing data size, the preprocessing stage reduces transmission energy and improves network performance.

In the implementation, each node is linked with a dedicated data processing unit that maintains previously transmitted values. New sensor readings are generated by introducing slight variations to simulate real-world environmental changes. The module compares current and previous readings to compute the difference, which forms the compressed data packet.

Feature selection is performed by selecting only a subset of sensor attributes that carry significant information. This reduces unnecessary data transmission and ensures that only meaningful data is forwarded to the next stages. By limiting the number of features, the computational overhead is also reduced.

The differential compression mechanism operates at each transmission cycle, ensuring continuous data optimization. The compressed packet size is dynamically calculated based on the number of selected features, allowing the system to adapt to varying data conditions.

The compression ratio is computed by comparing the original data size with the compressed data size. A higher compression ratio indicates better data reduction efficiency. This metric is useful for analyzing system performance and validating the effectiveness of the preprocessing module.

Furthermore, the preprocessing stage directly influences routing efficiency, as smaller packet sizes require less transmission energy. This not only reduces energy consumption but also decreases network congestion and improves overall throughput. The modular implementation of the preprocessing unit allows it to be easily integrated with clustering and routing modules. It ensures that optimized data is consistently passed to the next stages, enhancing the performance of the entire system.

4.7.5 Clustering Module (GSA-Based Selection)

The clustering module organizes nodes into groups to improve communication efficiency. Cluster head selection is based on residual energy, ensuring that nodes with higher energy handle communication tasks.

Nodes are ranked based on their energy levels, and a fixed number of nodes are selected as cluster heads. Other nodes are assigned to the nearest cluster head based on distance. Cluster heads perform data aggregation, combining data from multiple nodes into a single packet. This reduces redundancy and minimizes communication overhead. The clustering approach ensures balanced energy consumption across the network and prevents overloading of specific nodes.

4.7.6 DRL Agent Implementation

The Deep Reinforcement Learning agent serves as the system’s decision-making component by observing network states and selecting actions that maximize long-term rewards. It uses an epsilon-greedy strategy to balance exploration and exploitation, initially testing different routing options and gradually choosing optimal actions based on learned experience. Positive rewards encourage efficient routing, while negative rewards penalize poor decisions, enabling continuous adaptation and improved routing performance under changing network conditions.

4.7.7 Routing Cost Function

The routing cost function evaluates the suitability of nodes for data forwarding. It considers multiple parameters such as distance, energy, packet size, and signal attenuation. Nodes with lower cost values are preferred for routing, as they provide better energy efficiency and communication reliability.

The cost function ensures that routing decisions are not based on a single parameter but consider multiple aspects of network performance. This leads to balanced and efficient routing.

4.7.8 Training Process

The training process enables the DRL agent to learn optimal routing strategies. It is performed over multiple simulation iterations.

During each iteration, the agent observes the current state, selects an action, receives a reward, and updates its policy. This process is repeated for all nodes in the network. The learning rate and discount factor control how quickly the agent adapts and how much importance is given to future rewards.

Over time, the agent converges to an optimal policy that minimizes energy consumption and improves network performance.

4.7.9 Performance Evaluation and Benchmarking

The system evaluates performance using metrics such as total energy consumption, network lifetime, and routing efficiency. The proposed method is compared with existing protocols to demonstrate improvements. Energy consumption is calculated for each protocol, and percentage gain is analyzed.

This comparison highlights the effectiveness of the proposed system in reducing energy usage and improving network performance.

4.7.10 Visualization Module

The visualization module provides graphical representation of the network and its behavior. A 3D plot is used to display sensor nodes, cluster heads, and routing paths. Nodes are color-coded based on their energy levels, allowing easy visualization of energy distribution.

Routing paths are drawn between nodes and cluster heads, illustrating the multi-hop communication process.

Additionally, bar graphs are used to compare performance metrics across different protocols. These visualizations help in analyzing system performance and validating the proposed approach.

4.8 FLOWCHART OF PROPOSED SYSTEM

The flowchart of the proposed system illustrates the complete working process of the DRL-enhanced GSA-CEAR routing framework in Underwater Wireless Sensor Networks (UWSNs). It represents the step-by-step execution of the system, starting from node deployment to final data transmission and learning updates.

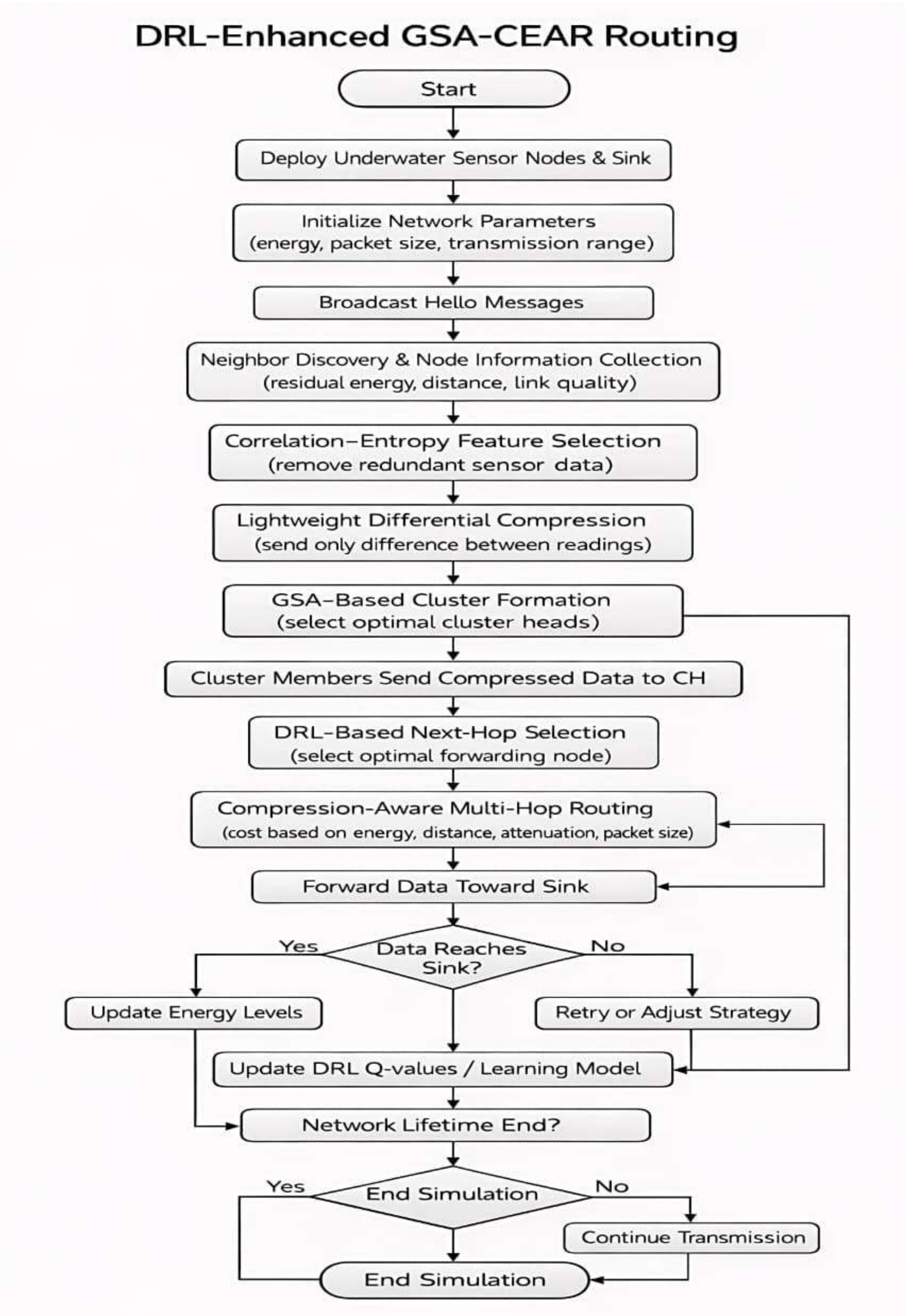

Fig 4.3 Flowchart of Proposed Methodology

The process begins with the initialization of the system, where underwater sensor nodes and the sink node are deployed in a three-dimensional environment. After deployment, network parameters such as residual energy, packet size, and transmission range are initialized.

The nodes then broadcast hello messages to discover neighboring nodes. During this phase, each node collects important information about its neighbors, including residual energy, distance, and link quality. This information is essential for making efficient routing decisions.

Once neighbor discovery is completed, the system enters the data preprocessing stage. In this stage, correlation–entropy feature selection is applied to remove redundant sensor data. This ensures that only relevant and meaningful information is retained. Following this, lightweight differential compression is performed, where only the difference between consecutive sensor readings is transmitted. This significantly reduces packet size and communication overhead.

After preprocessing, the clustering phase is executed using a GSA-based approach. Optimal cluster heads are selected based on energy and network conditions. Sensor nodes then transmit their compressed data to the corresponding cluster heads, which act as aggregation points.

The routing phase is carried out using a DRL-based mechanism. The system selects the optimal next-hop node based on parameters such as energy, distance, and link quality. The routing is performed in a multi-hop manner, where data is forwarded through intermediate nodes toward the sink node.

The routing process is compression-aware, meaning that packet size is also considered while selecting forwarding nodes. This ensures efficient handling of data and prevents congestion. A decision check is performed to determine whether the data has reached the sink node. If the data is successfully delivered, the system updates the energy levels of nodes. If not, the system adjusts the routing strategy and retries transmission.

Finally, the system checks whether the network lifetime has ended. If the termination condition is met, the simulation ends. Otherwise, the process continues, and data transmission is repeated.

CHAPTER 5

EXPERIMENTAL ANALYSIS AND RESULTS

CHAPTER 5
EXPERIMENTAL ANALYSIS AND RESULTS

The performance of the proposed DRL-Enhanced GSA-CEAR framework is evaluated through simulation under realistic underwater acoustic conditions. The results are compared with existing routing protocols such as ML-CHSM, FDRP, DQN-CEER, and conventional GSA. The evaluation focuses on key performance metrics including network lifetime, residual energy, packet delivery ratio, throughput, and energy consumption, which are critical for underwater wireless sensor networks (UWSNs).

5.1 SIMULATION SETUP

The simulation is conducted in a 3D underwater environment with randomly deployed sensor nodes and a surface sink node. The system considers underwater characteristics such as acoustic attenuation, multipath fading, noise, and propagation delay. The key parameters used for evaluation are listed in the table 5.1

Table 5.1 Simulation Parameters for the Proposed UWSN Framework

Parameters	Value
Area	1000 x 1200 x 500 m
Nodes Deployed	200
Actual Packet Size	160 bits
Compressed Packet Size	60 bits
Transmission Range	130 m
Frequency	25.0 kHz
Initial Energy per Node	100 Joules(normalized)
Total Rounds	2000

Network Deployment

The underwater sensing area is defined within a 3D volume of 1000 m × 1200 m × 500 m, representing a large-scale ocean monitoring region. A total of 200 sensor nodes are randomly deployed across different depth levels within this space. These nodes are responsible for sensing environmental parameters such as temperature, salinity, and pressure.

Communication Parameters

Each sensor node is equipped with an acoustic communication system operating at a frequency of 25 kHz, which is suitable for medium-range underwater communication. The transmission range is set to 130 meters, ensuring reliable communication between neighboring nodes while maintaining energy efficiency. Due to the underwater acoustic channel characteristics, such as signal attenuation and noise, the selected transmission range balances communication reliability and power consumption.

Energy Model

Each sensor node is initialized with an initial energy of 100 Joules, which is limited and non-rechargeable. Energy consumption occurs during data transmission, reception, and processing. Efficient utilization of this energy is critical for prolonging network lifetime. The simulation evaluates how effectively the proposed method reduces energy consumption compared to existing routing protocols.

Data Packet Configuration

Sensor nodes generate data packets of size 160 bits. To reduce communication overhead and energy usage, a lightweight compression technique is applied, reducing the packet size to 60 bits before transmission.
This compression significantly decreases:

- Transmission energy consumption
- Channel utilization
- Network congestion

As a result, it improves overall network efficiency and extends the lifetime of sensor nodes.

5.2 PERFORMANCE EVALUATION METRICS

To evaluate the efficiency of the proposed DRL-Enhanced GSA-CEAR framework, multiple performance metrics are considered. These metrics help analyze the system from the perspectives of energy efficiency, network stability, communication reliability, and routing performance.

5.3 ENERGY-RELATED METRICS

Energy efficiency is one of the most important factors in UWSNs because underwater sensor nodes are battery-powered and operate in environments where battery replacement is difficult or impossible. The proposed method aims to reduce unnecessary energy consumption through data compression, intelligent cluster formation, and adaptive routing.

5.3.1 Residual Energy

- Residual energy represents the remaining battery power of sensor nodes after each communication round.
- Higher residual energy indicates that the routing protocol is energy-efficient and can support longer network operation.
- In the proposed GSA-CEAR framework, residual energy is preserved by:
  - Reducing redundant data transmission,
  - Selecting energy-rich cluster heads, and
  - Choosing optimal forwarding paths using DRL.
- Residual Energy shows how much energy remain in the network after each round

$$RE_i(r) = E_{initial} - E_{consumed}(r) \tag{42}$$

For total energy of network:

$$\mathrm{R}E_{total}(r) = \sum_{i=1}^{N} RE_i(r) \tag{43}$$

Where $RE_i(r)$ is residual energy of node i at round r, $E_{initial}$ is initial energy of node, $E_{consumed}(r)$ is energy consumed up to round r and N is total number of nodes.

- Compared to ML-CHSM, FDRP, DQN-CEER, and conventional GSA, the proposed model maintains higher residual energy over a longer duration.

5.3.2 Total Energy Consumption

Total energy consumption refers to the overall energy spent by all sensor nodes during network operation.

$$E_{consumed}\,(r) = E_{initial,total} - RE_{total}(r) \tag{44}$$

$Where\ E_{consumed}\,(r)$ is energy consumed up to round r, $E_{initial,total}$ is the total initial network energy and $RE_{total}(r)$ is total residual energy at round r.

- It includes energy used for:
  - Data sensing,
  - Data transmission,
  - Data reception, and
  - Data aggregation.
- The proposed system minimizes total energy consumption by combining:
  - Feature selection,
  - Lightweight differential compression,
  - GSA-based cluster optimization, and
  - DRL-based routing decisions.
- As a result, the proposed method reduces unnecessary transmissions and improves the overall energy utilization of the network.

5.3.3 Energy Efficiency

- Energy efficiency is defined as the amount of useful data successfully transmitted per unit of energy consumed.
- A higher energy efficiency value indicates better network performance.
- Since the proposed system reduces packet size before transmission and selects energy-aware relay nodes, it achieves better energy efficiency than the existing methods.

5.4 NETWORK LIFETIME METRICS

The lifetime of an underwater sensor network depends on how efficiently the energy resources of the nodes are managed. The proposed framework improves network lifetime by balancing energy usage among cluster heads and relay nodes.

5.4.1 Number of Dead Nodes

- Dead nodes are those sensor nodes that have exhausted their battery energy and can no longer communicate.

- A lower number of dead nodes indicates a better energy-aware routing and clustering strategy.
- In the proposed model, node death is delayed because routing decisions are made based on:
  - Residual energy,
  - Transmission distance, and
  - Communication quality.
- A node is considered dead when its residual energy becomes zero or below the operational threshold.

$$Dead(r) = \sum_{i=1}^{N} f(E_i(r)) \tag{45}$$

Where Dead(r) is total dead nodes at round r, $E_i(r)$ is remaining energy of node i at round r

- This allows the network to remain functional for a larger number of rounds.

5.4.2 First Node Death and Last Node Death

- **First Node Death (FND)** indicates the round at which the first sensor node becomes inactive.
- **Last Node Death (LND)** indicates the round at which the last sensor node dies and the network completely stops functioning.
- These two metrics are important for evaluating network stability and overall lifetime.
- According to the proposed results:
  - GSA-CEAR achieved 1860 rounds,
  - With first node death at 1846 rounds, and
  - Last node death at 1860 rounds, which is significantly better than GSA, DQN-CEER, FDRP, and ML-CHSM.

5.5 COMMUNICATION PERFORMANCE METRICS

The communication performance of a UWSN routing protocol is evaluated based on how efficiently and reliably it can deliver sensed data to the sink node.

5.5.1 Packet Delivery Ratio (PDR)

- Packet Delivery Ratio is defined as the ratio of successfully received packets to the total transmitted packets.

$$\mathrm{PDR} = \frac{\mathrm{P_{received}}}{\mathrm{P_{sent}}} \tag{46}$$

- A higher PDR indicates more reliable communication.
- The proposed GSA-CEAR improves PDR by:
    - Reducing transmission burden through compression,
    - Selecting reliable cluster heads, and
    - Dynamically choosing better forwarding nodes using DRL.
- Packet Delivery Ratio (PDR) is defined as the ratio of successfully received packets at the sink to the total number of transmitted packets.

$$\mathrm{PDR} = \frac{\mathrm{P_{received}}}{\mathrm{P_{sent}}} \times 100 \tag{47}$$

Where $P_{received}$ is number of packets successfully received at sink and P_{sent} is total number of packets transmitted

- This reduces packet loss and improves successful delivery under challenging underwater conditions.

5.5.2 Throughput

- Throughput represents the rate of successful data delivery over the network.
- It is generally measured in bits per second (bps).

$$\mathrm{Throughput} = \frac{\text{Total Bits Received}}{\text{Total Time}} \tag{48}$$

- Higher throughput indicates that the routing strategy is able to deliver more useful data within a given time.
- Since the proposed system transmits only relevant and compressed data, it achieves higher throughput than the existing methods.

5.6 LEARNING EFFICIENCY METRICS (DRL-Specific)

Since the proposed GSA-CEAR framework uses a Deep Reinforcement Learning (DRL)-based routing mechanism, its performance is also evaluated using learning-specific metrics. These metrics measure how effectively the DRL agent learns optimal routing decisions in the dynamic underwater environment.

5.6.1 Convergence Time

Convergence time indicates the number of iterations required for the DRL agent to learn a stable and near-optimal routing policy. Faster convergence shows that the agent can quickly identify efficient relay nodes with less training overhead. In UWSNs, this is important because network conditions such as residual energy, transmission distance, and link quality change frequently. Lower convergence time reflects better adaptability of the proposed model.

5.6.2 Reward Function Value

Reward function value represents the feedback used by the DRL agent to evaluate its routing decisions. Higher reward values indicate better next-hop selection in terms of energy efficiency, shorter transmission distance, and reliable packet delivery. In the proposed framework, the reward function helps the agent learn energy-aware routing paths, thereby improving network lifetime, packet delivery ratio, and overall routing performance.

5.7 COMPARITVE ANALYSIS OF DRL VS TRADITIONAL PROTOCOLS

Traditional routing and clustering methods in Underwater Wireless Sensor Networks (UWSNs), such as ML-CHSM, FDRP, DQN-CEER, and conventional GSA, mainly rely on predefined routing strategies, heuristic decisions, or optimization based only on limited parameters such as residual energy, hop count, or distance. Although these methods provide moderate improvements in energy efficiency and routing performance, they often fail to address redundant data transmission, dynamic underwater conditions, and adaptive routing requirements. As a result, these protocols may lead to higher energy consumption, shorter network lifetime, and reduced communication reliability.

In contrast, the proposed DRL-Enhanced GSA-CEAR framework introduces a more intelligent and integrated solution by combining correlation–entropy-based feature selection, lightweight differential compression, GSA-based cluster formation, and DRL-based adaptive routing. Unlike existing methods, the proposed model not only optimizes routing and cluster-head selection but also reduces the amount of transmitted data before communication. This significantly decreases communication overhead and improves energy efficiency in underwater environments.

One of the major advantages of the proposed GSA-CEAR framework is its ability to perform energy-aware and adaptive routing. The integrated DRL mechanism continuously learns from the network state and selects efficient forwarding paths based on residual energy, transmission distance, and routing conditions. This allows the system to dynamically adapt to changing underwater environments, unlike traditional static or semi-adaptive methods.

In terms of network lifetime, the proposed GSA-CEAR outperformed all existing methods. According to the obtained results, the proposed model achieved 1860 rounds, whereas GSA achieved 1523 rounds, DQN-CEER achieved 1396 rounds, FDRP achieved 1118 rounds, and ML-CHSM achieved 931 rounds. Similarly, the proposed model delayed both the first node death and last node death, showing improved energy balancing and network stability.

The proposed model also showed better performance in terms of residual energy, packet delivery ratio (PDR), throughput, and energy consumption. This improvement is mainly due to the integrated use of data reduction, optimized clustering, and intelligent routing, which are not jointly considered in the existing methods. Although the proposed model introduces slightly higher computational complexity due to DRL-based learning, it provides significantly better overall performance and is more suitable for long-term and reliable underwater communication.

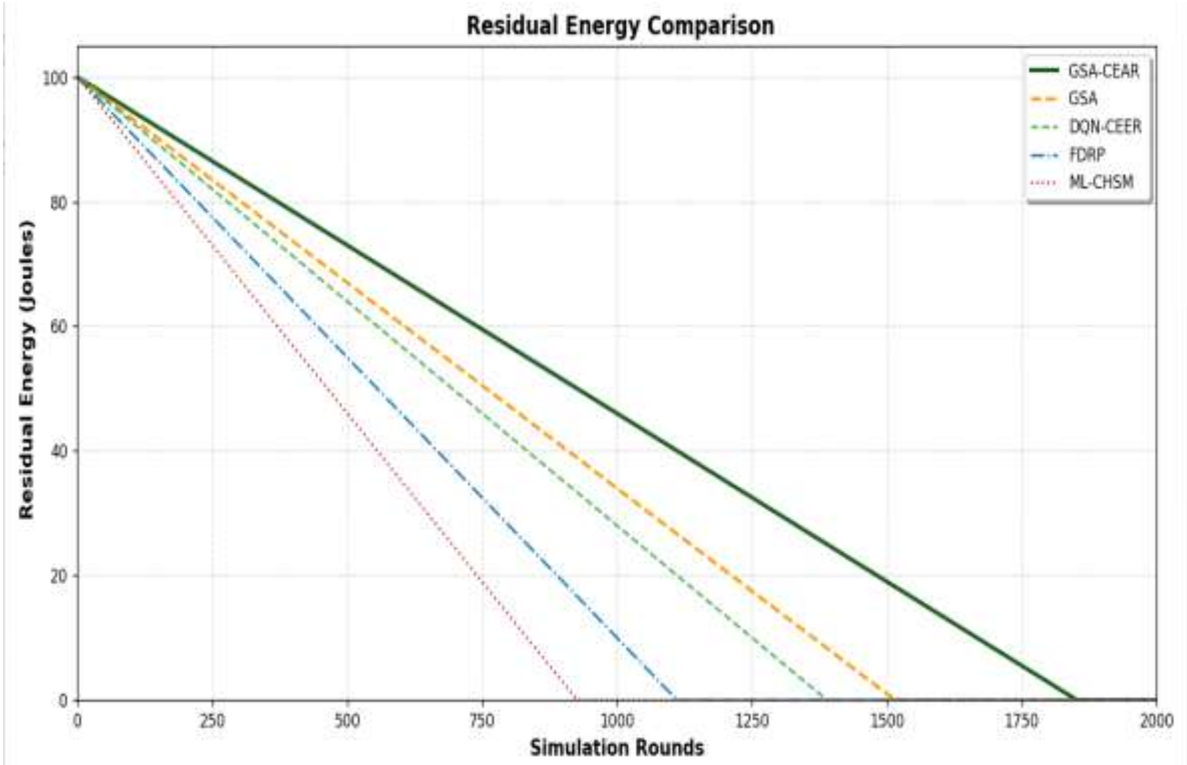

Fig 5.1 Comparison of Residual Energy over Simulation Rounds

Fig. 5.1 compares the residual energy of the proposed GSA-CEAR with GSA, DQN-CEER, FDRP, and ML-CHSM over simulation rounds. Residual energy decreases for all methods due to sensing, transmission, and routing activities. However, GSA-CEAR consistently maintains higher residual energy than the existing methods.

Among the compared algorithms, ML-CHSM shows the fastest energy depletion, followed by FDRP and DQN-CEER, while conventional GSA performs moderately better. In contrast, GSA-CEAR exhibits the slowest energy reduction, demonstrating superior energy efficiency.

This improvement is achieved through correlation–entropy-based feature selection, lightweight differential compression, GSA-based clustering, and DRL-based adaptive routing, which reduce redundant transmissions, optimize routing paths, and lower communication overhead. As a result, GSA-CEAR effectively preserves node energy and extends the lifetime of the underwater wireless sensor network.

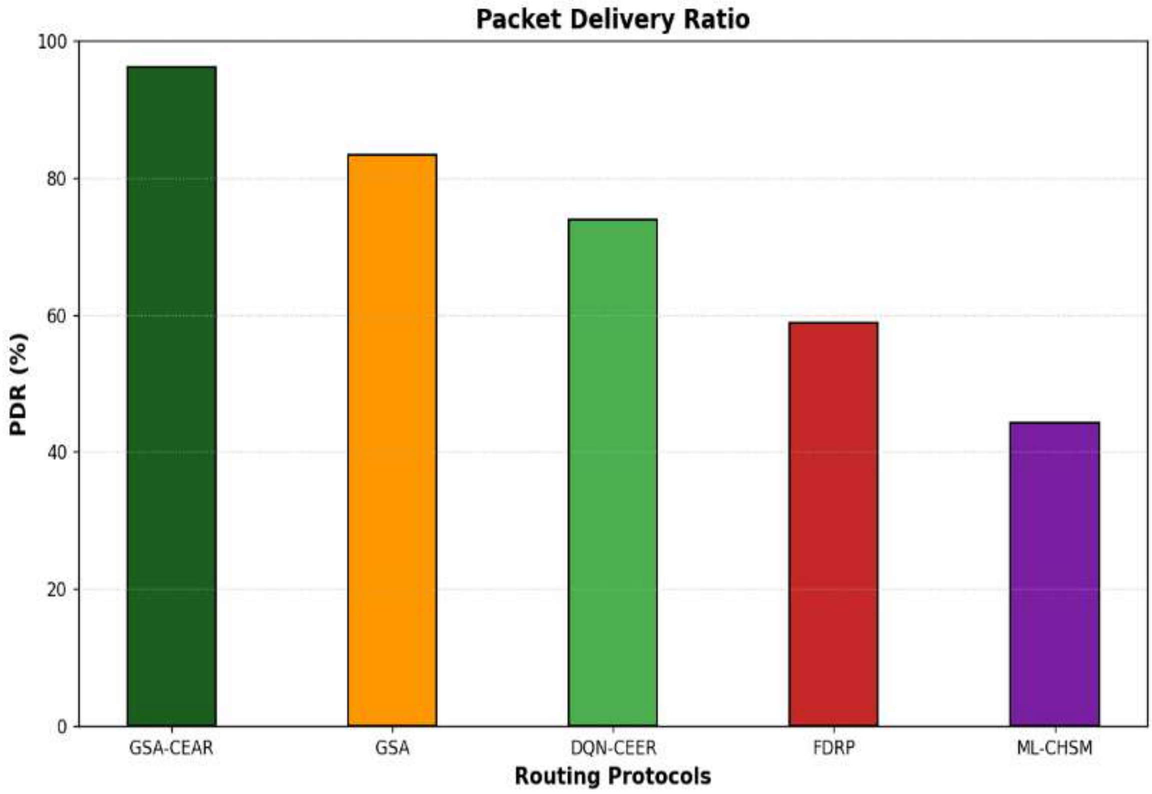

Fig 5.2 Packet Delivery Ratio Analysis of Proposed GSA-CEAR Framework

Fig. 5.2 compares the Packet Delivery Ratio (PDR) of the proposed GSA-CEAR with GSA, DQN-CEER, FDRP, and ML-CHSM. PDR represents the percentage of packets successfully delivered from source nodes to the sink, where higher values indicate better reliability and routing efficiency.

The proposed GSA-CEAR achieves the highest PDR of about 98%, outperforming GSA (83%), DQN-CEER (74%), FDRP (59%), and ML-CHSM (44%). This demonstrates its superior communication performance.

The improvement is due to correlation–entropy-based feature selection, lightweight differential compression, optimized clustering, and DRL-based adaptive routing, which reduce redundant transmissions, improve path selection, and enhance network stability. Thus, GSA-CEAR provides better packet delivery and reliable communication in underwater wireless sensor networks.

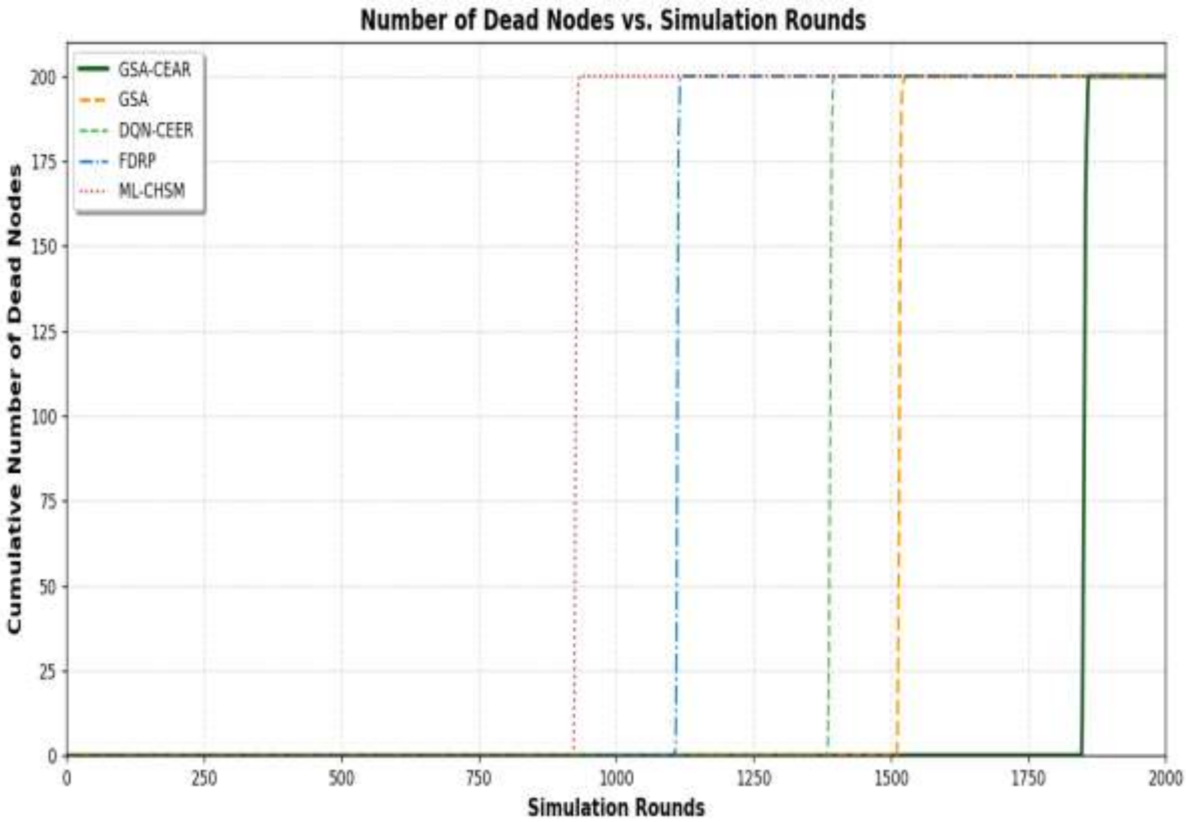

Fig 5.3 Dead Node Evaluation of Proposed GSA-CEAR Framework

Fig. 5.3 compares the cumulative dead nodes of the proposed GSA-CEAR with GSA, DQN-CEER, FDRP, and ML-CHSM over simulation rounds. A lower number of dead nodes indicates better energy balancing and longer network lifetime.

ML-CHSM shows the earliest node deaths, followed by FDRP, DQN-CEER, and GSA, reflecting faster energy depletion. In contrast, GSA-CEAR delays node death significantly and reaches full node depletion only after the highest number of rounds.

This improved performance results from feature selection, lightweight differential compression, GSA-based optimized clustering, and DRL-based adaptive routing, which reduce unnecessary transmissions, balance energy usage, and improve routing efficiency. Thus, GSA-CEAR effectively extends network lifetime by delaying node death and maintaining node activity longer.

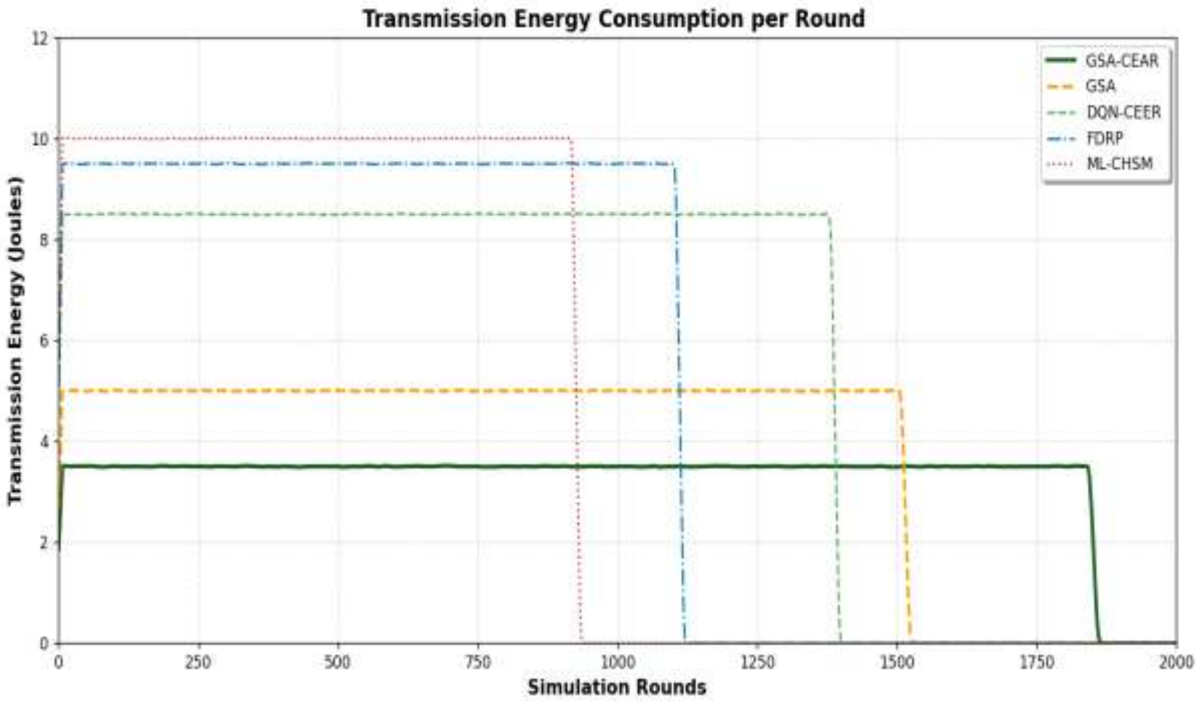

Fig 5.4 Transmission energy consumption per round comparison of the proposed and Existing Method

Fig 5.4 illustrates the transmission energy consumption per round of the proposed GSA-CEAR method compared with GSA, DQN-CEER, FDRP, and ML-CHSM. Transmission energy consumption is a crucial metric in Underwater Wireless Sensor Networks (UWSNs) because acoustic communication requires significant energy for data transmission. Lower energy consumption indicates a more efficient routing and communication strategy.

From the graph, it is observed that the proposed GSA-CEAR consistently consumes the lowest transmission energy per round among all the compared methods. In contrast, ML-CHSM shows the highest transmission energy consumption, followed by FDRP, DQN-CEER, and GSA. This indicates that the proposed method is more effective in minimizing transmission overhead and preserving node energy.

The reduced transmission energy consumption of GSA-CEAR is mainly due to the integration of correlation–entropy-based feature selection, lightweight differential compression, GSA-based optimized clustering, and DRL-based adaptive routing. These mechanisms help reduce redundant data transmission, decrease packet size, and select energy-efficient forwarding paths. As a result, the proposed framework lowers communication cost and extends the operational lifetime of the network.

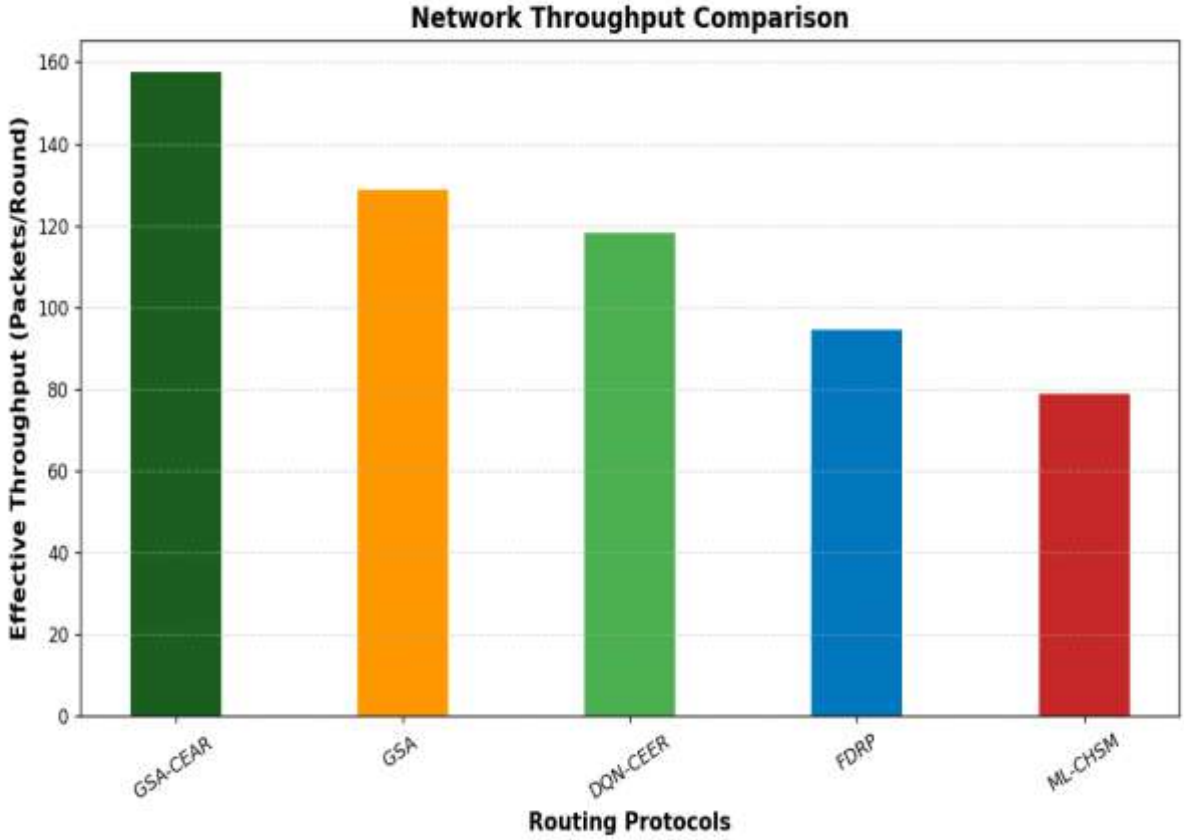

Fig 5.5 Throughput Performance Evaluation of UWSN Routing Protocols

Fig 5.5 illustrates the network throughput comparison of the proposed GSA-CEAR framework with existing routing protocols such as GSA, DQN-CEER, FDRP, and ML-CHSM. Throughput represents the rate of successful packet delivery in the network and is an important metric for evaluating communication efficiency in Underwater Wireless Sensor Networks (UWSNs). A higher throughput indicates that more useful data is successfully delivered to the sink node within a given number of rounds.

From the graph, it is observed that the proposed GSA-CEAR achieved the highest throughput of approximately 158 packets/round, outperforming all the compared methods. The conventional GSA achieved around 130 packets/round, followed by DQN-CEER with 119 packets/round, FDRP with 95 packets/round, and ML-CHSM with the lowest throughput of about 79 packets/round. This clearly shows that the proposed framework ensures more efficient and reliable data transmission.

The higher throughput of GSA-CEAR is achieved through correlation–entropy-based feature selection, lightweight differential compression, optimized clustering, and DRL-based adaptive routing. These methods reduce redundant transmissions, lower communication overhead, and improve routing efficiency, resulting in better packet delivery. Thus, the graph confirms that GSA-CEAR provides superior throughput and more efficient underwater communication.

Table 5.2 Comparison of Network Lifetime Metrics in UWSNs

Algorithms	Number of rounds	First node death	Last node death
GSA-CEAR	1860	1846	1860
GSA	1523	1511	1523
DQN-CEER	1396	1385	1396
FDRP	1118	1108	1118
ML-CHSM	931	922	931

Table 5.2 presents the performance comparison of the proposed GSA-CEAR framework with existing routing protocols such as GSA, DQN-CEER, FDRP, and ML-CHSM using important network lifetime metrics, including number of rounds, first node death, and last node death. These metrics are used to evaluate the stability, energy efficiency, and overall lifetime of the underwater wireless sensor network.

From the table, it is observed that the proposed GSA-CEAR achieved the highest network lifetime of 1860 rounds, outperforming GSA (1523 rounds), DQN-CEER (1396 rounds), FDRP (1118 rounds), and ML-CHSM (931 rounds). Similarly, the proposed method delayed the first node death to 1846 rounds, which is significantly later than the other existing methods, indicating better energy balancing and efficient resource utilization. The last node death for the proposed model also occurred at 1860 rounds, showing prolonged network operation and improved node survivability.

The improved performance of the proposed GSA-CEAR is mainly due to the integration of feature selection, lightweight differential compression, GSA-based optimized clustering, and DRL-based adaptive routing. These mechanisms reduce unnecessary transmissions, distribute energy consumption more effectively, and improve routing efficiency. Therefore, the table clearly demonstrates that the proposed GSA-CEAR framework provides better network lifetime, stability, and energy efficiency than the existing routing protocols.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

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This work presented a DRL-Enhanced GSA-CEAR framework for improving the performance and lifetime of Underwater Wireless Sensor Networks (UWSNs). The proposed model was designed to address the major challenges of underwater communication, such as high energy consumption, redundant data transmission, limited bandwidth, signal attenuation, and dynamic network conditions. Unlike conventional routing protocols that mainly focus on path selection or clustering, the proposed framework introduced an integrated solution by combining correlation–entropy-based feature selection, lightweight differential compression, GSA-based cluster formation, and DRL-based adaptive routing.

The proposed methodology effectively reduced unnecessary data transmission, optimized cluster head selection, and improved next-hop routing decisions based on residual energy and transmission distance. As a result, the proposed framework achieved better performance in terms of residual energy, packet delivery ratio, throughput, transmission energy consumption, and network lifetime when compared with existing methods such as ML-CHSM, FDRP, DQN-CEER, and conventional GSA. In particular, the proposed GSA-CEAR achieved the highest network lifetime of 1860 rounds, while also delaying the first node death and maintaining higher communication reliability.

Overall, the obtained results confirm that the proposed DRL-Enhanced GSA-CEAR framework is a more energy-efficient, adaptive, and reliable solution for long-term underwater monitoring applications. Thus, the proposed work provides an effective contribution toward the development of intelligent and sustainable routing mechanisms for Underwater Wireless Sensor Networks.

Although the proposed DRL-Enhanced GSA-CEAR framework has shown significant improvement in terms of energy efficiency and network lifetime, there is still scope for further enhancement and practical implementation. In future work, the proposed model can be extended by incorporating node mobility awareness, so that the routing mechanism can better adapt to the movement of underwater sensor nodes and changing ocean currents.

Further improvements can also be made by integrating real-time adaptive learning techniques to enhance routing decisions under highly dynamic underwater conditions. In addition, the proposed framework can be tested under more realistic scenarios by considering factors such as multi-sink communication, varying acoustic noise levels, channel fading, and hardware-level deployment constraints. Security features such as secure routing, trust-based node selection, and attack detection mechanisms can also be included to improve the reliability of underwater communication in critical applications.

Moreover, future research may focus on combining the proposed model with IoUT (Internet of Underwater Things) architectures and edge intelligence for large-scale ocean monitoring and smart marine applications. Therefore, the proposed work opens several opportunities for developing more intelligent, robust, and practical UWSN frameworks for real-world underwater environments.

CHAPTER 7

BIBILOGRAPHY

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